

Style and Skill: Hedge Funds, Mutual Funds, and Momentum

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Abstract

Classifying mandatory 13F stockholding filings by manager type reveals that hedge fund strategies are mostly contrarian, while mutual fund strategies are largely trend following. The only institutional performers—the 2/3 of hedge fund managers that are contrarian—earn alpha of 2.4% per year. Contrarian hedge fund managers tend to trade profitably with all other manager types, especially when purchasing stocks from momentum-oriented hedge and mutual fund managers. Superior contrarian hedge fund performance exhibits persistence and stems from stock-picking ability rather than liquidity provision. Aggregate short sales further support these conclusions about the style and skill of various fund manager types.

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The high returns from buying assets that are past winners and selling past losers represents one of the most prominent and studied asset pricing anomalies in finance.¹ “Momentum,” as it is frequently called, is simple to implement and popular, particularly among institutional investors. For example, Grinblatt, Titman, and Wermers (1995) and Carhart (1997) show that mutual funds tend to follow momentum strategies. Prior research also shows that momentum accounts for almost all of the measured stock-selection ability of mutual fund managers: Once momentum is controlled for, (pre-expense) alpha is negligible or disappears entirely. Grinblatt and Keloharju (2000) find that Finnish institutions, mostly mutual funds, are also momentum investors and earn superior returns.

Despite widespread research exploring how past returns influence mutual fund managers’ trading styles, little evidence exists on whether momentum trading is also favored by what many regard as the most sophisticated and incentivized managers: those who run the three trillion dollar hedge fund industry.² Motivated by this fact, we develop an extensive hedge fund manager database from a comprehensive sample of 13F institutional stockholdings. This database of investment advisers’ quarterly stock positions allows us to analyze the investment style and performance of 589 mutual fund, 1,342 hedge fund, and 2,894 other institutional fund managers from 1998 to 2012, with mutual fund managers alone overseeing more than 2,600 mutual funds.

Prior research on hedge fund style generally employs return databases, which are contaminated by expenses, asset marks that smooth hedge fund returns, as well as survivorship, backfill, and reporting biases. Moreover, studies of asset manager trading styles typically analyze earlier sample periods pre-dating the modern era in which there are thousands of hedge funds, or use limited numbers of funds or fund indexes constructed from limited numbers of funds. For example, Grinblatt, Titman, and Wermers (1995) study asset manager behavior only until 1984, while Agarwal and Naik (2004) analyze hedge fund index risk exposures before 2002, finding that eleven of twelve hedge fund return indices, each representing a hedge fund strategy class, exhibit insignificant exposure to the Carhart (1997) momentum factor, with one strategy, relative value arbitrage, appearing significantly contrarian. Recently, Jylhä, Rinne, and Suominen (2014) use a regression weighting of prior daily returns to produce a liquidity factor, finding that hedge funds provide liquidity. Jame (2018) examines transaction data for 70 hedge fund advisers. Like Jylhä, Rinne, and Suominen (2014), he finds that

¹See Jegadeesh and Titman (1993), Fama and French (2008), and Avramov, Chordia, Jostova, and Philipov (2013).

²Assets under management as of the 2nd quarter of 2018 according to the online source BarclayHedge: https://www.barclayhedge.com/research/indices/ghs/mum/HF_Money_Under_Management.html.

hedge funds offer liquidity-supplying trades. However, this contrarian behavior is short-run and says little about contrarian trading reactions to returns from less recent days.³

We document that most hedge fund managers are not momentum traders. For almost 2/3 of hedge fund managers, stock purchases tend to be contrarian, although their tendency to sell past winners is less pronounced. Moreover, contrarian hedge fund managers' trading style is persistent: About 80% of hedge fund managers that are contrarian in the first half of the sample period are also contrarian in the second half. Consistent with prior study of mutual fund style, we find that about 2/3 of mutual fund managers follow momentum strategies, both for purchases and sales. Their investment style is persistent: Conditional on their survival, 78% of mutual fund advisers that follow momentum strategies in the first half of the sample period also follow the same strategies in the second half. Our style results are robust to differences in turnover and are unlikely to be affected by consideration of short sales or intra quarter trades, which are omitted from the 13F data.

Previous papers on hedge fund performance also tend to rely on self-reported fund returns, finding mixed results. One camp finds that hedge funds with greater managerial incentives earn superior returns (Agarwal, Daniel, and Naik 2009) and that the performance of top hedge fund managers persists and cannot be explained by luck (Kosowski, Naik, and Teo 2007, Jagannathan, Malakhov, and Novikov 2010). The other camp documents that fewer than 20% of long/short equity hedge funds consistently deliver alpha (Fung and Hsieh 2001) and hedge funds on average outperform mutual funds, but not the market, while being more volatile than both mutual funds and the market. Furthermore, questions remain about the reliability of self-reported returns: Patton, Ramadorai, and Streatfield (2015) provide recent evidence that returns in hedge fund databases are frequently revised; the revisions are predictable and tend to increase average returns by 0.62% per month.⁴

We find strong evidence of hedge fund adviser performance, as well as performance persistence only among outperforming hedge fund managers: Underperforming hedge fund managers and outperforming or underperforming managers of mutual funds and other institutional investment vehicles show no performance persistence. There are significant alphas only for the 2/3 of hedge fund advisers who are contrarians. Contrarian hedge fund advisers not only outperform all benchmarks,

³The contrarian strategies studied in this paper condition on past returns and differ from book-to-market-conditioned value strategies. For example, what Lakonishok, Shleifer, and Vishny (1994) call "contrarian" (trading against glamour) differs from what we call contrarian, a style that relates trading to 3-month past returns. The cross-sectional correlation between past returns and book-to-market ratios is low and positive (0.02).

⁴Although Agarwal, Jiang, Tang, and Yang (2013) observe that SEC filings may also be revised, the revisions—being infrequent and motivated by private information—cannot be viewed as attempts to manipulate performance.

but also mutual fund advisers and the approximately 1/3 of hedge fund managers classified as momentum traders. Contrarian hedge fund managers' success derives from their superior stock-picking skills—the ability to pick the *right* past losers among stocks with similar characteristics.

In contrast to hedge funds, momentum mutual fund managers outperform contrarian mutual fund managers. However, once we control for the return enhancing effects of momentum, performance differences between the two styles of mutual fund management disappear. Our hedge fund performance results are robust to different risk-adjustment methodologies and are not driven by survivorship bias, distorted marks, cross-subsidization within fund families, and specific subperiods. We also provide evidence that the potential observability of short sales would only enhance our performance findings.

The style measure we use, first proposed by Grinblatt, Titman, and Wermers (1995) [GTW], is based on the (vector) inner product of the fund's portfolio weight changes with past stock returns. This product is positive when stocks with relatively high past returns subsequently witness portfolio weight increases and those with lower past returns generate portfolio weight decreases—indicating a momentum strategy. Our measure of fund adviser performance is the analogous product, but with future returns replacing past returns. Thus, performance measures the ability of the manager's portfolio weight changes to predict future returns, following Grinblatt and Titman (1993) [GT]. An equivalent perspective is that the GT measure evaluates active management by comparing adviser returns to a benchmark return that would be earned if the adviser maintained prior stockholdings.

We further adjust the GT performance measure for the impact of stock characteristics. In particular, the performance measure uses returns that are size- and book-to-market-adjusted as in Daniel, Grinblatt, Titman, and Wermers (1997) [DGTW]. Where appropriate, we also adjust for momentum with the same DGTW procedure. The adjusted performance measure tests for true stock-picking skills by adding an extra level of assurance—that a managed portfolio is not beating its benchmark (the passive portfolio it replaced) in a given month because it deviates from that portfolio in its size, book-to-market, and momentum attributes. The GT performance measures reveal that mutual fund managers do not beat benchmarks that adjust for momentum, whereas hedge fund advisers outperform their benchmarks irrespective of whether they contain a momentum control. The superior hedge fund performance is persistent and entirely attributable to the contrarian subgroup of hedge fund managers. Contrarian hedge fund managers beat their passive benchmark by 2.4% per year based on size, book-to market, and momentum-adjusted returns, with two-thirds of this average

fund performance coming from the purchases of stocks.

Our holdings-based evaluation of investment style and performance has several advantages over comparable analyses of reported fund returns. First, holdings-based hedge fund evaluation avoids well-known biases in hedge fund return databases—including survivorship, backfill, and self-selection biases (see [Fung and Hsieh, 1997, 2002b](#))—and possible misreporting (see [Patton, Ramadorai, and Streatfield, 2015](#)). These biases arise because reporting to a hedge fund return database is voluntary and hedge funds can choose to report (or stop reporting) following strong (poor) performance, or revise their reports in a self-serving manner. Our sample does not suffer from these biases because we estimate fund performance using quarterly holdings reported in *mandatory* 13F SEC filings and stock returns from CRSP. Second, the granularity of fund holdings allows us to distinguish active investment decisions (changes in quantities held) from passive momentum investing (due to changes in prices) when evaluating a fund manager’s strategy and performance. Third, we can compute holdings-based measures of style and performance without the contaminating effects of incentive fees and high-water marks, which can introduce non-linearity and path dependence in hedge funds’ returns (see [Getmansky, Lo, and Makarov, 2004](#)). Finally, the fund manager’s prior-quarter holdings offer a benchmark for style and performance evaluation. Choosing the right benchmark is especially important for hedge fund advisers, which hold stocks with different characteristics than other institutional investors ([Griffin and Xu, 2009](#)) and follow strategies with option-like payoffs ([Fung and Hsieh 1997, 2001, 2002a,b, Mitchell and Pulvino 2001, Agarwal and Naik 2004](#)).

While we are not the first to use fund holdings data, even for hedge funds, our analysis uses a much larger cross-section and longer post-1997 time series to provide novel findings on hedge fund style and performance. Portions of the 13F filings have been used before to study hedge fund performance. For example, [Brunnermeier and Nagel \(2004\)](#) analyze the 13F holdings of 53 hedge fund managers during the 1998-2000 technology bubble. [Griffin and Xu \(2009\)](#), using a representative sample of 306 hedge fund advisers from 1980-2004, find no ability to forecast returns from the levels of or the changes in 13F hedge fund holdings, although large bets have some forecasting power for future returns.⁵ These earlier studies derive useful insights from hedge fund portfolio holdings. However, comprehensive

⁵Other research using 13F stockholdings includes [Ben-David, Franzoni, and Moussawi \(2012\)](#), who study drivers of the drastic decrease in hedge fund holdings in 2008, and [Agarwal, Jiang, Tang, and Yang \(2013\)](#), who study the “confidential holdings” of hedge fund advisers (permitted for certain information-sensitive holdings and reported with delay) and show that these have superior performance. Recently, [Sias, Turtle, and Zykaj \(2016\)](#) find that hedge fund managers drive prices toward fundamental values rather than destabilize prices by crowding out equity trades.

and bias-free examination of the hedge fund industry with pre-1998 data poses challenges because hedge fund databases systematically retained records on dead funds only since 1998.

We conduct additional tests at the stock level to confirm the results at the fund adviser level, examine the interactions between different types of fund managers, and analyze the effects of liquidity provision, seasonalities, and short selling activity. Consistent with the results at the fund adviser level, we find that stocks that are purchased by contrarian hedge fund advisers have an average DGTW-adjusted alpha of 2.7% per year and outperform the stocks sold by these advisers (excluding short sales) by an average alpha of 3.4% per year. We also look at fund-level mutual fund holdings from the Thomson Reuters S12 database, obtaining results that are consistent with those based on the 13F adviser-level holdings.

Hedge fund managers' success as contrarian investors is particularly large for contrarian hedge fund buys of stocks that momentum mutual funds sell. In this case, the former managers earn alphas of 4.2% per year while the latter have a negative alpha of the same magnitude. Our evidence is consistent with the claim that contrarian hedge fund advisers are likely on the correct side of the trade in their interactions with other fund manager types. Moreover, when trading alongside contrarian hedge fund advisers, the adviser is always better off when agreeing on a stock's trade direction (buy vs. sell) with contrarian hedge funds than when disagreeing.

To better assess the issue of stock-picking ability vs. liquidity provision, we analyze stock-level performance using residuals from cross-sectional regressions. The similarity of these results to DGTW stock-level performance, the insensitivity of the results to controls for liquidity effects, and the seasonal pattern of performance with this alternative methodology suggest that the performance of contrarian hedge fund advisers is unlikely to stem from liquidity provision. The seasonal pattern also could imply that other investors mimic the reported trades of hedge funds and that the measured performance of hedge fund managers would be larger if more frequent data on trades were available.

These results are further reinforced by the analysis of aggregate short positions. Selling stocks that contrarian hedge fund managers purchase is a particularly perilous strategy for stocks with below-median short interest, which are among the most profitable trades of contrarian hedge fund advisers. Purchases of stocks with below-median levels of short interest earn contrarian hedge fund advisers alphas of 5.4% per year, while for above-median short interest the comparable average stock alpha is 2.5% per year. To the extent that short interest in a stock is representative of hedge fund

positions, greater short interest contributes positively to the performance of hedge funds observed from their long positions. Overall, these findings show that contrarian hedge fund advisers exhibit superior stock-picking ability that goes beyond a naïve style-based strategy or liquidity provision.

In sum, this paper offers four main contributions to the literature. First, it represents the first large cross-section that documents contrarian hedge fund behavior at past return horizons that are too long to be interpreted as liquidity provision. Second, it shows that contrarian hedge fund managers are the only fund managers exhibiting stock-picking skills. Third, we identify performance persistence among outperforming hedge fund managers. Finally, our holdings data empirically confirm that hedge fund advisers profitably buy stocks that mutual funds later regret they sold.

1. Data

Institutional stock ownership is computed from the mandatory quarterly 13F filings of investment advisers. Since 1980, all institutional investment advisers managing over \$100 million must report their stockholdings on Form 13F to the Securities and Exchange Commission. Reporting of end-of March, June, September, and December holdings is required for all long stock positions of 10,000 or more shares, as well as any position valued at over \$200,000. The first 13F filing is due within 45 days after the end of the calendar year in which the adviser meets the \$100 million filing threshold. Rule 13f-1(a)(1) also requires filings for at least 3 quarters after falling below the \$100 million threshold. There is no exemption for bankruptcies and liquidation as long as any reportable securities remain in the adviser's portfolio. These reporting requirements, along with the use of performance measures that track holdings-implied returns for at least one quarter after each adviser's last report was filed, ensure that the sample is free of survivorship bias.

The 13F filings do not report the types of funds the adviser manages. We classify each 13F filing as belonging to a mutual fund adviser, hedge fund adviser, or other adviser. Mutual fund advisers are identified from the Thomson Reuters S12 mutual fund holding database, which provides their advisers' corresponding number on the 13F file. The remaining 13F filers are manually identified as hedge fund or other advisers by matching their names with a list of hedge fund manager names—comprehensive since 1998—obtained from various sources, including Lipper TASS, Hedge Fund Research (HFR), Center for International Securities and Derivatives Markets (CISDM), Morningstar, Bloomberg, and Businessweek Private Company List. In addition, we cross-check our adviser

classification against Form ADV for all advisory firms registered under the Investment Advisers Act.⁶ When an ADV is filed, a firm is classified as a hedge fund adviser only if the form indicates that more than 50% of its regulatory assets under management (AUM) belong to pooled investment vehicles other than investment companies (a category including hedge funds and other private funds).⁷

Using these procedures, no firm we classify as a hedge fund adviser has any mutual funds under management. However, about 1 in 5 of the advisers classified as mutual fund advisers have a small portion of their AUM from hedge funds. After separating the pure mutual fund advisers from the hybrid fund advisers, who manage both mutual and hedge funds, we find that the style and performance of the two mutual fund adviser types is similar, as reported in Internet appendix Tables A1 and A2. These adviser-level results are also verified using fund-level data from the Thomson Reuters S12 mutual fund holdings database, which are comprehensive and mandatory for mutual funds (see Internet appendix Table A3).

The 13F reports have the limitation of omitting intra-quarter positions, short positions, derivatives, and small positions. Filers may also request confidentiality, with sufficient justification, for up to a year on some of their positions.⁸ Focusing exclusively on non-negligible long equity positions has the advantage of facilitating comparisons of ability across managers. For instance, our observed differences in style and skill do not reflect hedge fund managers' access to more complex trading instruments or to fancy dynamic trading strategies that synthetically generate derivatives with nonlinear returns and complex risk adjustments. While style and skill measures computed from quarterly long equity positions are not affected by these limitations, they offer an incomplete portrait of managerial style and skill that might be evident from the missing holdings. The omission of derivatives positions and confidential holdings makes it less likely that we will identify superior performance among skilled hedge fund advisers who frequently make use of these tools. However, this would make any findings of skill conservative. Supporting this conclusion is our later analysis of the impact of short sales, obtained by supplementing the 13F data with aggregate short positions when analyzing style and performance at the stock level.

⁶Prior to 2011, many hedge fund advisers relied on the private fund adviser exemption to avoid registration under the Advisers Act. Most advisers who qualified for the exemption are hedge fund advisers. We classify all unregistered advisers matched with a hedge fund manager name as hedge fund managers because these advisers could not receive an exemption and manage any fund marketed to non-qualified investors (including any mutual fund). From 2011, the Dodd-Frank Act eliminated the ADV filing exemption for hedge fund advisers with regulatory assets over \$150 million.

⁷The ADV form does not provide information on the percentage of assets attributable to each type of client prior to 2011. We use form ADV's information on the percentage of clients of each type instead.

⁸Agarwal, Jiang, Tang, and Yang (2013) provide an excellent analysis of confidential 13F filings.

Over our 1998-2012 sample period, we identify 589 unique mutual fund advisers, 1,342 hedge fund advisers, and 2,894 “other” advisers. Prior to 1998, we cannot reliably identify hedge fund managers, because the databases used to identify hedge fund managers do not retain dead funds until 1998. Plot A of Figure 1 presents the number of mutual and hedge fund advisers in each quarter. The number of mutual fund advisers is fairly stable, ranging between 299 and 379; the number of hedge fund advisers, by contrast, increases over the sample period, starting at 165 in March 1998, reaching a maximum of 804 in June 2008, and finishing at 650 hedge fund advisers in December 2012. The number of “other” advisers also increases over the period. Overall, the sample contains 21,219 mutual fund, 29,050 hedge fund, and 73,821 other adviser-quarter 13F filings (Panel A of Table 1).

Mutual fund advisers hold larger portfolios—on average 374 stocks, compared to 128 for hedge fund advisers and 233 for other fund managers (Panel B of Table 1 and Plot B of Figure 1). The average dollar value of stockholdings is \$11.29 billion per mutual fund adviser with \$30.42 million per stock. This compares to \$864 million per hedge fund adviser with \$6.80 million per stock, and \$3.1 billion for others with \$13.23 million per stock. Hedge fund managers have twice the turnover of mutual funds—39.2% versus 19.7% (see also Plot C of Figure 1).⁹

Mutual and hedge fund managers fundamentally differ in the types of stocks they hold (Panel C of Table 1). Relative to mutual funds, hedge fund managers hold stocks with *higher* returns (0.90% vs. 0.65% per month), book-to-market ratios (0.68 vs. 0.57), credit risk (rated BBB vs. A–), volatility, analyst EPS forecast dispersion (0.21 vs. 0.14), and short interest ratios (4.58% vs. 3.24%), but with *lower* prices (\$26.95 vs. \$33.84), market capitalizations (\$22 vs. \$41 billion), trading volumes (\$1.59 vs. \$2 million per month), analyst coverage (14 vs. 18 analysts), earnings surprises (SUE of 0.61 vs. 0.85), and dividends (\$0.38 vs. \$0.62 per share). Stocks held by the two types of managers are similar in terms of bid-ask spreads¹⁰ (0.38% vs. 0.36%) and institutional ownership (0.72% vs. 0.70%).

Panel D of Table 1 displays the overlap between mutual and hedge fund holdings. Of the stocks that hedge fund advisers hold, 99% are also held by mutual fund advisers. Yet, only 76% of the stocks held by mutual funds are also held by hedge funds. This may be because hedge fund managers prefer to pursue substantially more concentrated strategies than mutual fund advisers.

⁹Turnover is the average dollar values of buys and sells over the quarter relative to the initial holdings [i.e. $\text{turnover} = (\$ \text{buys}_q + \$ \text{sells}_q) / (2 \times \$ \text{holdings}_{q-1})$].

¹⁰Bid-ask spreads are computed using the monthly CRSP stock files as the difference between the ask and the bid prices divided by the price, which is an average of the two.

2. Methodology

2.1. Measuring Style

Following GTW, we assess whether an adviser follows a momentum strategy from Lag 0 Momentum (L0M), the vector product of (active) portfolio weight changes (observed quarterly) and past returns (observed monthly):

$$\text{L0M}_{iq} = \sum_{j=1}^{N(q)} (w_{i,j,q} - w_{i,j,q-1}) R_{j,q} \quad (1)$$

where L0M_{iq} is the L0M measure of fund manager i in quarter q , $w_{i,j,q}$ is fund manager's i 's quarter- q ending weight on stock j ,¹¹ $N(q)$ is the number of stocks in quarter q , and $R_{j,q}$ is the sum of stock j 's monthly returns in quarter q .¹² A positive L0M indicates a tendency towards momentum investing—the fund manager is buying stocks with positive past returns and/or selling stocks with negative past returns. Conversely, a negative L0M points to contrarian investing.

Each quarter, we average the L0M_{iq} measures across fund advisers to evaluate their aggregate tendency (L0M_q) to follow momentum. The average style measure (L0M) along with its sample t -statistic is obtained from the time series of these aggregate tendencies:

$$\text{L0M}_q = \frac{1}{K(q)} \sum_{i=1}^{K(q)} \text{L0M}_{iq} \quad (3)$$

$$\text{L0M} = \frac{1}{Q} \sum_{q=1}^Q \text{L0M}_q \quad (4)$$

$$t\text{-stat}(\text{L0M}) = \frac{\text{L0M}}{s.e.(\text{L0M})} = \frac{\text{L0M}}{\sigma(\text{L0M}_q)/\sqrt{Q}} \quad (5)$$

where $K(q)$ is the number of fund managers of a given type (hedge, mutual, or other) in quarter q and Q is the number of quarters in the sample.

We also compute L0M using a cross-sectional average in lieu of a time series average. In this case, we first obtain an average L0M_i for each fund adviser i over the number of quarters, $Q(i)$, she

¹¹To insure that the measure captures only “active” (i.e. discretionary) weight changes caused by changes in quantities rather than prices, we use the prior quarter price, $p_{j,q-1}$, when computing weight changes:

$$w_{j,q} - w_{j,q-1} = \frac{\text{shares held}_{j,q} \times p_{j,q-1}}{\sum_{j=1}^N \text{shares held}_{j,q} \times p_{j,q-1}} - \frac{\text{shares held}_{j,q-1} \times p_{j,q-1}}{\sum_{j=1}^N (q) \text{shares held}_{j,q-1} \times p_{j,q-1}} \quad (2)$$

¹²We use sums of the quarter's 3 monthly returns (or non-cumulative quarterly return) instead of the cumulative quarterly return for consistency with Grinblatt, Titman, and Wermers (1995).

files a 13F, and then average across the total number, K , of fund managers of a given type:

$$\text{L0M}_i = \frac{1}{Q(i)} \sum_{q=1}^{Q(i)} \text{L0M}_{iq} \quad (6)$$

$$\text{L0M} = \frac{1}{K} \sum_{i=1}^K \text{L0M}_i \quad (7)$$

Compared to the time series average, the cross-sectional average places more weight on shorter tenure managers and those operating in months with greater numbers of advisers. While the time series t -statistic in equation (5) is a standard means test for i.i.d. returns, no similar independence exists for the fund adviser-level components of the cross-sectional average. As the analogous t -statistic would be biased, we omit test statistics when reporting the cross-sectional average.

A fund's investment style may differ on the buy and sell sides or may be influenced more by one side than the other. To capture this difference, we measure an adviser's buy and sell momentum style, respectively L0Mb and L0Ms, from quarterly measures of the style as follows:

$$\text{L0Mb}_{iq} = \sum_{w_{i,j,q} > w_{j,q-1}} (w_{i,j,q} - w_{i,j,q-1})(R_{j,q} - B_{i,q}) \quad (8)$$

$$\text{L0Ms}_{iq} = \sum_{w_{i,j,q} < w_{j,q-1}} (w_{i,j,q} - w_{i,j,q-1})(R_{j,q} - B_{i,q}), \quad (9)$$

where $B_{i,q}$ is a benchmark return, a constant for each fund adviser i in quarter q , which equals the stockholdings-weighted average of the returns of the size and book-to-market portfolios to which each stock j belongs,

$$B_{i,q} = \sum_{j=1}^{N(q)} w_{i,j,q-1} \text{SZBM}_{j,q}, \quad (10)$$

where $\text{SZBM}_{j,q}$ is the quarter q return of stock j 's size- and book-to-market-matched portfolio, computed as in Daniel, Grinblatt, Titman, and Wermers (1997).¹³ The benchmark return in equation (10) is a proxy for the *fund adviser's* expected return that quarter based on the size and book-to-market characteristics of the fund's beginning-of-quarter portfolio.

While equation (1)'s measure of investment style is zero under the null hypothesis of no past

¹³Specifically, each month, NYSE, AMEX, and Nasdaq stocks are sorted into value-weighted portfolios based on a sequential 5×5 sort on size and book-to-market, using NYSE breakpoints. Size is measured as of the end of the prior June and book-to-market is as of the end of the prior fiscal year. The size and book-to-market portfolios to which the stock belongs are based on its past rankings of the two characteristics. The benchmark's quarterly return is the sum of the three montly returns.

return influence on trades, the buy and sell style measures based on raw returns lack this property. To address this issue, the buy and sell style measures in equations (8)-(9) subtract an *adviser-specific* benchmark return in each quarter. Without this adjustment, the buy measure will have a positive bias because weight changes and returns tend to be positive—spuriously indicating momentum—whereas the sell measure will have a negative bias. Since weight changes add to zero across purchases and sales, subtracting that fund-specific benchmark has no effect on the overall LOM measure. The benchmark subtraction in equations (8) and (9) is purely designed to ensure that the buy and sell versions of the measure properly estimate the covariance between weight changes and past returns.

2.2. Measuring Performance

We evaluate fund performance as the vector product of the fund manager's (active) weight changes and future risk-adjusted returns. This performance measure, adapted from Grinblatt and Titman (1993), employs the portfolio held by the manager at the end of the previous quarter as its benchmark,

$$F1M_{iq}^* = \sum_{j=1}^N (w_{i,j,q} - w_{i,j,q-1}) R_{j,q+1}^*, \quad (11)$$

where $R_{j,q+1}^*$ is the non-cumulative risk-adjusted return for stock j in quarter $q + 1$. A measure greater than zero indicates that purchases outperform sales. Thus, the F1M performance measure is much like the LOM style measure except that a) it uses next quarter's future returns rather than lagged returns and b) adjusts returns for risk. Moreover, future returns are risk-adjusted relative to a benchmark which ideally assigns zero performance to all passive portfolios. As Grinblatt and Titman (1989, 1993) show, active managers can only achieve alpha-generating performance by timing portfolio weights to hold more future winners and fewer future losers. Hence selling stocks with negative future risk-adjusted returns or buying stocks with positive future risk-adjusted returns increases F1M. By contrast, poorly timed trades, like selling stocks with positive future risk-adjusted returns, decrease F1M.

Following DGTW, the stock-by-stock risk adjustment for F1M* matches each stock in a given month to a value-weighted portfolio of stocks in the same quintiles for both size and book-to-market ($* = SzBm$) or for size, book-to-market, and momentum ($* = DGTW$). The risk adjustment consists of subtracting the return of the matched portfolio from the return of the stock.

Even without a risk adjustment for the return of each stock held, the expected value of the F1M measure is zero under the null hypothesis of no investment skill, provided that, over the next quarter, the portfolio of stocks recently purchased by the manager tend to have the same priced risk as the stocks sold. We risk adjust returns because the priced risk of a stock may be dynamic and managers may dispose of stocks when their risk has declined in favor of stocks with higher priced risk. Hedge fund managers, for example, tend to hold smaller stocks with higher book-to-market ratios (see Table 1 Panel C). Such stocks tend to have positive market-adjusted returns due to the size and value effects, but the size and value attributes of a stock tend to change over time.¹⁴

Unlike L0M, there is no need to further adjust F1M* if we want to separately compute the performance contribution from purchases or sales of assets. Returns are already risk adjusted. However, in contrast to L0Mb and L0Ms, this adjustment takes place at the stock level rather than the fund adviser level. To obtain the separate performance contributions of buys and sells, referred to as F1Mb*_{iq} and F1Ms*_{iq}, we simply sum the terms of equation (11) separately for weight changes that are positive and negative, respectively. We compute time series and cross-sectional averages of this performance measure with the same aggregation sequences used for L0M. We also perform quarterly Fama-MacBeth cross-sectional regressions of fund-level measures, F1M_{iq}, on dummy variables indicating whether the adviser manages mutual (MF) or hedge funds (HF).

3. Results

3.1. Hedge Fund and Mutual Fund Adviser Styles

Table 2 presents our style measure. For each type of adviser, Panel A presents the time series mean of the cross-sectional average L0M (as in equation 4) with *t*-statistics in parentheses (as in equation 5). The *t*-statistics test whether time series of (non-overlapping) portfolio returns that generate L0M have a mean of zero. Square-bracketed numbers represent the fraction of quarters with positive style measures.

Table 2 Panel A reveals that hedge fund managers are contrarian, while, consistent with Grinblatt, Titman, and Wermers (1995), mutual fund managers are momentum investors. Mutual funds' quarterly L0M measures average 0.71 (*t*=14.88). That is, the prior-quarter returns of stocks mutual

¹⁴A comparison with measures based on raw returns is provided in the Robustness section.

funds buy exceed the prior-quarter returns of stocks they sell by 71 basis points. The measure is positive in every quarter of the sample period. Mutual fund managers follow momentum both in their buys and sells. Their buys' average L0Mb style measure averages 0.39 (positive in 87% of the quarters) and their sells' L0Ms measure averages 0.32 (positive in 82% of the quarters). By contrast, hedge fund advisers tend to be contrarians. Their L0M measure averages -0.34 ($t = -3.62$) and is negative in 70% of the quarters. Collectively, hedge fund advisers are contrarian only in their buys, having an average L0Mb of -0.37 , indicating a significant tendency to buy past losers. However, their insignificant L0Ms of 0.03 is partly an artifact of style heterogeneity across hedge fund managers. As Internet appendix Table A4 shows, contrarian hedge fund advisers (those with $L0M < 0$) are contrarian on both the buy and sell side while momentum hedge fund advisers ($L0M > 0$) are momentum traders on both the buy and sell side. Note, though, that momentum hedge fund advisers' sales exhibit a more intense style tilt towards momentum than the corresponding contrarian style tilt of the contrarian hedge fund advisers' sales. This makes Table 2's style measure for the pooled sales of the two hedge fund style-types appear relatively neutral with respect to past returns.¹⁵

Panel A of Table 2 also computes the cross-sectional average of L0M, L0Mb, and L0Ms. It first averages the measures across all quarters for each adviser, and then across all advisers of the same type. These averages confirm that mutual fund managers have a strong tendency to buy recent winners and sell losers whereas hedge funds tend to buy recent losers. Differences from the time series averages are due to the different weighting schemes: The cross-sectional average weights all funds equally; the time series average places relatively greater weight on advisers with longer tenures in the sample period. We do not report t -statistics for the cross-sectional average because of potential bias due to correlations among advisers with the same style.¹⁶

Figure 2 illustrates how the mutual and hedge fund advisers' L0M style measures evolve over time. Plot A demonstrates consistent momentum trading by mutual fund advisers throughout the sample period, while hedge fund advisers exhibit contrarianism, particularly since 2000. Indeed,

¹⁵A relatively less intense focus on sales is consistent with prior studies documenting an asymmetric information content of buys and sells. For example, Grinblatt, Keloharju, and Linnainmaa (2012) find that smart investors' stock purchases are more informed than sales while Saar (2001) proposes a theoretical model in which the permanent price impact of buys is larger than that of sells.

¹⁶If we could observe intra-quarter round-trip trades, it is theoretically possible that the measured styles of various manager types could be altered. To further address this concern, we compute style measures separately for advisers with below- and above-medium turnover, as measured from quarterly holdings. As Internet appendix Table A5 shows, both high- and low-turnover hedge fund advisers are strongly and about equally contrarian. High-turnover mutual fund advisers tend to exhibit greater momentum style than their low-turnover counterparts, likely reflecting the dilution arising from lack of change in many of the positions of low-turnover mutual funds.

hedge fund advisers are relatively more contrarian than mutual funds in all but one quarter. Hedge funds' contrarian trading was particularly prescient just before and during stock market meltdowns—the dot-com bubble burst in 2001 and the 2008 financial crisis. Hedge funds' switched from a fairly neutral to a strong contrarian stance in the second half of 2000 and became far more contrarian between the end of 2006 and the end of 2008. By contrast, mutual fund advisers intensified their momentum trading between mid-2007 and mid-2008, but there is little evidence of other time-related complementarities in style between mutual and hedge funds. Indeed, over the whole sample period, the correlation between mutual and hedge fund advisers' average trading styles is insignificantly positive (0.29). The insignificant correlation suggests that mutual fund trading is not a major determinant of variation in hedge funds' trading style over time.

Plots B and C of Figure 2 separately show the buy and sell LOM measures, LOMb and LOMs, for the two adviser types. During the 2008-2009 financial crisis and the 2001-2002 dot-com bubble burst, hedge fund advisers intensified their buys and sells of losing stocks.¹⁷ However, the increased buying of losers outweighed the increased selling of these stocks, making hedge funds' overall strategy more contrarian during the two meltdown periods. Mutual fund advisers' buy and sell LOM measures are typically both positive, indicating a persistent tendency to buy winners and sell losers.

The bottom section of Panel A of Table 2 shows the distribution of momentum and contrarian advisers by fund type. About two-thirds of hedge fund advisers are contrarian and two-thirds of mutual fund advisers are momentum traders. While more than two-thirds of the hedge fund advisers are contrarian in their buys, only about half are contrarian in their sells. In contrast, most mutual fund advisers follow momentum in both their buys and sells.

Panel B of Table 2 documents that most advisers demonstrate persistent investment styles. After sorting fund advisers into momentum and contrarian investors based on their LOM measure in the first half of the period (from January 1998 to June 2005), Panel B of Table 2 reports the average LOM and proportion of contrarian advisers (negative LOM) during the second subperiod (July 2005 to December 2012) for various fund types, conditional on their continued existence in both subperiods. As can be seen in Panel B, 79% of hedge fund advisers that are contrarian in the first subperiod remain contrarian in the second; 64% ($=100\%-36\%$) of the momentum hedge fund managers from the first half also follow momentum in the second half. Similarly, 78% ($=100\%-22\%$)

¹⁷Hedge fund advisers exhibited a momentum trading style in several quarters during the 1998-99 price run-up of technology stocks. Their change in style is consistent with findings in Brunnermeier and Nagel (2004).

of the momentum mutual fund advisers from the first half also follow momentum in the second, and 71% of the contrarian mutual fund advisers from the first half are also contrarian in the second half.

3.2. Performance of Mutual and Hedge Fund Advisers

We now examine the F1M performance measure (equation 11), which is based on the covariance between active weight changes and subsequent stock returns. We compute two versions of the performance measure that control for changing risk premia in individual stocks by using risk-adjusted returns. The first, $F1M^{SzBm}$, controls for size- and book-to-market [BM]; the second, $F1M^{DGTW}$, controls for size, book-to-market, and momentum. Table 3 Panel A reports time series averages of the cross-sectional means of the F1M measures of mutual and hedge funds with t -statistics in parentheses, and the fraction of months in which the cross-sectional means are positive in square brackets.

Panel A shows that mutual fund managers generate 13 basis points per quarter [bpq] ($t=2.67$) from active changes in their holdings based on size- and BM-adjusted returns. In contrast, active changes in hedge funds' holdings earn 27 bpq ($t=3.61$) more over the subsequent quarter than stocks with similar size and BM ($F1M^{SzBm}$). The comparison is even less favorable to mutual fund advisers once we control for momentum: The $F1M^{DGTW}$ measure shows that mutual fund managers generate an insignificant 2 bpq after adjusting performance for the return-enhancing effects of their momentum investment style. That is, mutual fund managers perform as well as a naïve momentum investor. Mutual fund managers' buy and sell performance measures are also insignificant, irrespective of controls, and negligible in magnitude (below 4 bpq) once we control for momentum. By contrast, controlling for their contrarian style has little effect on hedge fund manager performance: Their $F1M^{DGTW}$ measure of 25 bpq ($t=3.72$) negligibly differs from the 27 bpq performance metric that lacks the past return control.¹⁸

The superior performance of hedge fund advisers derives entirely from stock purchases, which earn 28 bpq (SzBm-adjusted) and 26 bpq (DGTW-adjusted), respectively, both significant at the 1% level. Since the performance measures are based on characteristic-adjusted returns, hedge fund performance derives from their advisers' stock-picking ability rather than from following a strategy

¹⁸Momentum premium adjustments for stocks bought and sold by hedge funds have little effect because (as Internet appendix Table A6 shows and we later discuss) hedge fund managers measure winners and losers only from returns in the last quarter. By contrast momentum mutual fund advisers are also momentum traders at horizons that precede the prior quarter. The winners minus losers momentum premium is relatively small for strategies formulated from returns in the past three months and larger when winners and losers are defined by returns from the past six months.

based on the size, value, or momentum anomalies. While hedge funds' buys are contrarian, their advisers earn alpha by picking the 'right' losers, in that these stocks outperform other past losers with similar size and BM.

The performance measures at the bottom of Panel A—calculated as the cross-sectional average of the fund-specific F1M measures—give equal weight to each fund adviser, regardless of the adviser's longevity during the sample period. The cross-sectional average shows even stronger stock-picking skills for hedge fund managers than the time series averages. For example, the cross-sectional average $F1M^{SzBm}$ measure shows performance of 46 bpq—compared to the time series average of 27 bpq. The difference is driven by the greater hedge fund performance observed in the “sells” column. Thus, at least for the cross-sectional equal weighting of advisers, hedge fund managers derive a small portion of their performance from selling stocks that will soon earn inferior returns. By contrast, the mutual fund advisers' cross-sectional DGTW alphas in Table 3 are smaller than their time series alphas.

Overall, the performance measures in Panel A of Table 3 show that mutual fund managers' quarterly portfolio revisions yield DGTW alphas ranging from -8 bpq to 2 bpq, while hedge fund managers' portfolio revisions generate alpha from 25 to 40 bpq. The hedge fund managers' portfolio revisions remain profitable after netting out transaction costs. A back-of-the-envelope calculation has hedge fund managers' 39.2% turnover per quarter (from Panel B of Table 1), at an average round-trip cost of 0.38% (the bid-ask spread from Panel C of Table 1), generating total transaction costs of 15 bpq (mutual fund turnover, 19.7% per quarter, at a round-trip cost of 0.36%, implies total transaction costs of 7 bpq). The round-trip transaction costs per dollar of trade, estimated as the share-weighted average of each stock's CRSP-reported half bid-ask spread, are conservative because many transactions occur at prices inside the quoted spread (see [Hasbrouck, 2009](#)).

Figure 3 depicts a 3-year moving average of quarterly performance over time for mutual fund and hedge fund advisers. Plot A focuses on $F1M^{SzBm}$, Plot B on $F1M^{DGTW}$. In both panels, hedge funds earn abnormal style-adjusted returns and outperform mutual funds except for a brief period before the 2008 financial crisis. While mutual fund advisers almost always perform well based on size- and BM-adjusted returns in Plot A, Plot B shows that this performance is entirely driven by their momentum style. Controlling for momentum, the $F1M^{DGTW}$ measure reveals that mutual fund managers' performance is relatively flat and indistinguishable from zero. By contrast, controlling for past returns (Plot B) has virtually no effect on hedge fund managers' superior performance.

Figure 3 Plot C displays the separate DGTW-adjusted performance contributions of the stock purchases and sales of hedge fund advisers. It confirms the results from Table 3 that hedge fund advisers' superior performance is driven mostly by stock purchases. Hedge funds' sales positively contributed to their performance during the second half of 2008, exactly when hedge funds' buy performance was at its weakest. The stocks sold by hedge funds during the 2008 crisis subsequently underperformed their size, book-to-market, and momentum benchmark, indicating some degree of sell-side stock-picking skill during the crisis.

Panel B of Table 3 examines performance persistence. Specifically, it sorts mutual fund, hedge fund, and other advisers into two groups—underperformers and outperformers—based on the sign of their $F1M^{SzBm}$ or $F1M^{DGTW}$ measures during the first half of the sample period. It then reports their performance measures during the second half of the sample period. We find performance persistence among successful hedge funds but not among other types of fund advisers or poorly performing hedge funds. The hedge fund advisers that are successful in the first half of the sample period earn significant risk-adjusted returns of 43 bpq during the second half based on $F1M^{SzBm}$, and 33 bpq based on $F1M^{DGTW}$.

3.3. Performance and Investment Style

To assess whether investment style influences performance, Table 4 sorts fund advisers into contrarians and momentum subgroups based on the sign of their average L0M measure over the sample period.¹⁹ For each subgroup, we compute the $F1M^{SzBm}$ and $F1M^{DGTW}$ performance measures for the entire sample. Panel A reports the time series average of the quarterly performance measures for each subgroup; Panel B the corresponding cross-sectional averages.

Table 4 reveals a striking difference between the performance of contrarian and momentum hedge fund advisers. Only contrarian hedge fund advisers (the 867 advisers with $L0M < 0$ in Table 2 Panel B) achieve significantly positive alpha. Contrarian hedge fund managers outperform their size-, BM-, and momentum-adjusted benchmarks by 34 bpq in Panel A and 60 bpq in Panel B. The former figure significantly exceeds the corresponding alpha of momentum hedge funds, which is close to zero. (Panel B's cross-sectional averages do not easily lend themselves to statistical tests.)

¹⁹Our contrarian vs. momentum classification is based on the entire sample period. To verify that past performance does not drive later style choices, we also classified fund advisers into the two style-types using the first half of the sample period only, and the results are virtually the same.

The performance difference between contrarian and momentum hedge fund advisers is driven mostly by buys. Contrarian hedge funds' buys beat their DGTW style-adjusted benchmark by 33 bpq (Panel A) and 41 bpq (Panel B), while their sells outperform by only 1 bpq (Panel A) and 19 bpq (Panel B), respectively. In contrast, the average alphas of the 475 momentum hedge fund advisers are statistically indistinguishable from zero. Thus, the significant hedge fund performance observed in Table 3 is entirely driven by contrarian hedge fund advisers; they buy stocks that outperform their benchmarks in the subsequent quarter, and, based on Panel B's cross-sectional results, also sell stock they own that tends to underperform. Regardless of style, Table 4 shows no evidence of investment skill among mutual fund managers after adjusting for the effect of momentum on stock returns.²⁰

While Table 1 revealed substantial differences between hedge fund and mutual fund advisers, Internet appendix Table A7 shows that there are no fundamental differences between contrarian and momentum hedge fund managers' portfolios that might account for the former's superior performance—compared to momentum hedge funds, advisers of contrarian funds hold 7 fewer stocks, \$272 million (about 25%) less in AUM (the difference from mutual funds is more than tenfold), \$1.7 million less per stock, and have about 5% less turnover. Compared to all other types of advisers, while contrarian hedge fund managers choose stocks with higher volatility and higher credit risk, smaller earnings surprises (SUE), lower dividends, higher short interest ratios, and greater forecast dispersion than other adviser types, these stock attributes are associated with lower not higher returns. Moreover, we largely control for these and other attributes by using the prior quarter's holdings (adjusted for the DGTW characteristics) as a benchmark for computing alpha. The same table does indicate that contrarian hedge fund managers oversee portfolios with both fewer number of stocks and fewer dollars being managed. Perhaps these attributes make contrarian hedge fund advisers more flexible and better able to capitalize on superior information in a way that does not dilute performance by revealing that information before all trades are completed. To investigate this hypothesis, Internet appendix Table A8 runs quarterly Fama-MacBeth regressions of $F1M^{DGTW}$ on

²⁰To further test whether the intensity of momentum or contrarian behavior is rewarded, we study coefficients from Fama-McBeth regressions of a fund's quarter $q+1$ performance (both overall and for its buys and sells) on its quarter q L0M style measure (or the comparable L0M measure for its buys and sells). The results (unreported) are consistent with those reported in Table 4. These regressions, run separately for mutual and hedge fund advisers, show a significant positive relation between the L0M style measure and subsequent returns for mutual funds, but only if the returns do not control for stock momentum. The intercepts from the mutual fund regressions are insignificant, confirming that mutual fund managers' returns are an artifact of their momentum investing. By contrast, hedge funds' significant intercepts and insignificant slope coefficients—regardless of the benchmark—confirm that their performance is due to superior stock-picking skills, and not driven by their investment style.

the market capitalization of the adviser's holdings, the number of stocks held, adviser turnover, and dummies for whether the adviser type is contrarian hedge fund, momentum hedge fund, contrarian mutual fund, and momentum mutual fund (with "Other institutional adviser" as the omitted dummy). Panel A reports the average of the regression coefficients with only the dummies for adviser type as regressors, while Panel B reports the regression with both these dummies and the three other attribute controls. There is little change in the coefficients on the dummies for adviser type when the three controls are included.²¹ Hence, the source of contrarian hedge fund manager's superior stock-picking skill remains elusive.

4. Evidence from Stock Returns

Up to this point, the paper focused its performance analysis at the fund manager level. An alternative perspective comes from looking at the subsequent performance of the stocks that are bought or sold by various groups of fund advisers. This approach allows us to study the returns of stocks sorted both by the trades of groups of advisers and by trades between groups of advisers.

4.1. DGTW-Adjusted Stock Returns Sorted by Who Buys and Sells

We construct the buy (sell) portfolios for different types of advisers as follows: First, we classify advisers into groups based on their fund type (mutual or hedge fund) and style (momentum or contrarian, according to their average LOM_{*i*} over the sample period). We then compute the *monthly* alphas (here DGTW-adjusted returns)²² of an equally weighted portfolio of those stocks (with share prices exceeding one dollar and data on short interest) for which the aggregate shareholdings of a given group of advisers increased (the group's "buy" portfolio) or decreased (the "sell" portfolio) over the previous quarter. We also examine the alphas of equal-weighted portfolios of stocks that are net buys (or sells) of one adviser group and net buys (or sells) for another adviser group.

Table 5 reports the alphas of the buy, sell, and buy minus sell portfolios for each group of

²¹Internet appendix Table A8 also shows results using Newey and West (1987) adjusted *t*-statistics as well as clustered errors both by adviser and by time using the method of Thompson (2011). Most results are the same. Those that use Thomson's procedure are more significant.

²²Results are highly similar if we take the alphas from regressions of the DGTW-adjusted returns on the four factors of Carhart (1997).

advisers, as well as buy-sell pairings of groups of advisers.²³ Panel A shows mutual and hedge funds portfolios by adviser type and style; Panel B reports portfolios based on pairs of adviser subgroups. For example, contrarian hedge fund advisers and momentum mutual fund advisers have separate aggregate portfolio weight vectors that change over each quarter. Stocks that are bought by contrarian hedge fund advisers and sold by momentum mutual fund advisers (1st row and 2nd column of Panel B) are stocks for which the former fund category's aggregate portfolio weights increased and the latter category's weights decreased (after adjustment for the passive portfolio weight change due to stock price changes). The equal weighted portfolio return in the first row and second column of Panel B thus represents the average return of stocks that effectively are sold by momentum mutual fund advisers to contrarian hedge fund advisers.

Consistent with the paper's earlier adviser-level results, Panel A shows that a) the alphas of stocks bought by hedge fund advisers exceed those of the stocks they sold by 18 basis points per month [bpm], b) the 18 bpm difference stems from the buys of stocks that beat their DGTW benchmarks, and c) the alpha difference is due to the trades of contrarian hedge fund advisers. By contrast, stock sales of the mutual fund adviser group are cause for regret: These stocks tend to subsequently achieve large abnormal returns in the quarter after a sale is recorded, earning almost 21 bpm more than their DGTW benchmark.

Figure 4 shows that these findings are persistent. It graphs 36-month moving averages of the DGTW-adjusted returns for each of the four adviser type-style buy-minus-sell portfolios in Table 5 Panel A. Plot A shows the performance over time of the buy-minus-sell portfolio of contrarian advisers, whereas Plot B shows the performance of momentum advisers. The solid line in Plot A confirms that the trades of contrarian hedge fund advisers consistently produce a positive alpha over the sample period, while the solid line in Plot B shows that the trades of momentum hedge funds advisers do not generate alpha. The dashed lines also show consistent negative alphas for the mutual fund advisers' buy-minus-sell portfolios, irrespective of whether their strategy is contrarian (Plot A) or trend following (Plot B). Plot C illustrates that, over a 36-month period, contrarian hedge fund managers almost always outperform momentum hedge fund managers.

²³Note that, consistent with prior literature, a positively signed sell portfolio alpha implies bad fund performance, and vice versa. With stock-level analysis, the sign convention requires that we subtract sell from buy performance figures to obtain overall performance. For adviser-level F1M performance, bad sell performance was indicated by negative performance figures, a sign convention that is consistent with prior literature using the F1M measure. The different sign convention required buy and sell F1M performance to be added to obtain an overall alpha.

Table 5 Panel B has six 2x2 arrays that examine whether stock returns are influenced by concordance (each array's diagonal elements) or disagreement (off-diagonal elements) about the proper trade direction (buy or sell) between pairings of adviser type-style groups. Among pairings with contrarian hedge funds—the top two rows of alphas in Panel B—the contrarian hedge fund advisers are on the correct side of the trade in all 12 cases. The highest alpha stocks (35.3 bpm) are contrarian hedge fund purchases of stocks that momentum mutual fund advisers sell; the next highest (31.1 bpm) are contrarian hedge fund purchases that momentum hedge funds sell. In both cases, these tend to be losing stocks that the contrarian hedge fund managers believe are going to bounce back, while the momentum mutual and hedge fund managers likely view these as “value traps.” By contrast, in the middle two rows of Panel B, momentum hedge fund managers are on the correct side of the trade for their buys, irrespective of whether they are in accord or disagreement with any style of mutual fund. However, momentum hedge (and mutual fund) managers are always on the wrong side of the trade with their sells unless contrarian hedge fund managers also sell the same stocks.

Thus, Table 5's evidence indicates that the smart investors are the contrarian hedge fund advisers and that their performance is driven more by their purchases. To the extent that these contrarian hedge fund managers are purchasing what are true bargain stocks at the expense of other institutional investors, the greatest bargains are in stocks sold by momentum mutual and hedge funds.²⁴

4.2. Liquidity Provision

Selling pressures by momentum investors can cause temporary price distortions which benefit contrarian investors who step in and provide liquidity. This liquidity provision may generate alphas which could erroneously be attributed to superior stock selection. If the alphas of contrarian hedge fund advisers were due to liquidity provision, rather than superior stock selection, the returns of stocks traded by their funds would tend to show short-term price reversals centered around the trade dates. We address this possibility by further adjusting each stock's return for the effect of its prior-month return. The adjustment is similar to the DGTW size, BM, and momentum adjustment. However, instead of a four-way sort, which would introduce too much statistical noise due to the small number of stocks in each “bucket” of multi-dimensional characteristics, we implement cross-sectional regressions of stock returns on the four characteristics— $\log(\text{size})$, $\log(\text{BM})$, returns over months $t - 7$

²⁴These results are consistent with Massa, Simonov, and Yan (2012) who argue that hedge funds take advantage of mutual funds' constraints by showing that hedge funds perform better where mutual funds ownership is higher.

to $t - 2$, and month $t - 1$ returns. Table 6 (which mirrors Table 5) reports time series averages (or alphas) of the monthly cross-sectional means of the regression residuals for a given category. The alphas from this cross-sectional residual approach (henceforth, “CS-alphas”) are similar—both to CS-alphas from regressions without a 1-month past return adjustment (Table 7) and to our DGTW alphas in Table 5. To illustrate, in Panel A of these tables, contrarian hedge fund buys generate monthly CS-alphas of 23.6 (Table 6) and 24.2 bpm (Table 7), respectively. This finding suggests that liquidity provision does not substantially affect our conclusions about the stock-picking skills of contrarian hedge funds. Results (reported in Internet appendix Table A9) remain virtually unchanged when the DGTW-adjusted portfolio returns from Table 6 are further risk-adjusted with additional factors, including the Fama and French (2015) five factors, the Carhart (1997) momentum factor, the short-term reversal factor, the Hou, Xue, and Zhang (2015) q-model factors, the Stambaugh and Yuan (2017) mispricing factors, the Pástor and Stambaugh (2003) liquidity factor, the default factor, the Jylhä, Rinne, and Suominen (2014) and Nagel (2012) liquidity provision factors, and a short-interest ratio factor.²⁵

4.3. Seasonalities in Hedge Fund Performance

The superior performance of contrarian investors could also concentrate in January and perhaps be due to an anomaly tied to seasonal stock return reversals rather than superior stock-selection ability. The CS-alphas from Table 6, which control for past return reversals, suggest this hypothesis is unlikely. To further investigate this issue, Table 8 reports the spread in the CS-alphas of the stocks bought and sold by contrarian (Panel A) and momentum (Panel B) hedge fund advisers for each month of the year. Additional spacing in the table’s two panels separates calendar quarters, identifying when the reported portfolio weights change. For the contrarian hedge fund advisers of Panel A, the January alphas of stocks purchased significantly exceed the alphas of the stocks they sell; however, that same alpha spread (far right column) is larger in five of the remaining eleven

²⁵ The Fama and French (2015) five factors, the Carhart (1997) momentum factor, the short-term reversal factor are taken from Professor Kenneth French’s website: http://mba.tuck.dartmouth.edu/pages/faculty/ken.french/data_library.html. The Hou, Xue, and Zhang (2015) q-model factors are obtained from the authors. The Stambaugh and Yuan (2017) mispricing factors (<http://finance.wharton.upenn.edu/~stambaugh/>) and Nagel (2012) liquidity provision factor (<https://voices.uchicago.edu/stefannagel/code-and-data/>) are obtained from the authors’ websites. The Pástor and Stambaugh (2003) liquidity factor is from WRDS. The default factor is the return differential between BBB and AAA, from the Federal Reserve Bank of St. Louis: <https://fred.stlouisfed.org>. We compute the Jylhä, Rinne, and Suominen (2014) liquidity provision factor, following the methodology described in their paper, as the return differential of stock portfolios sorted on a weighting of past 20-day returns. We compute the short-interest ratio factor as the month- t return differential between the highest and lowest short-interest stock deciles as of month $t - 1$.

months. Note that the Panel A alpha spreads are smaller and always insignificant in the third month of each calendar quarter but the alphas in the first and second month of the calendar quarter tend to be similar. Because the last months of each calendar quarter (March, June, September, and December) represent months when the reported changes in portfolio holdings are most stale, this pattern is consistent with stock picking ability rather than seasonal reversals or liquidity provision.²⁶ Indeed, the pattern may indicate that the performance we report for contrarian hedge fund advisers is conservative and diluted only by the fact that changes in adviser holdings are reported quarterly rather than more frequently. For example, the contrarian hedge funds' buy-minus-sell alpha spread for the eight months that exclude March, June, September, and December sum to more than 400 basis points, which is about 6% per year annualized. The drop in Panel A's alpha spreads in the four months tied to the most stale holdings changes could be due to dissemination (and subsequent irrelevance) of the private information that triggered the contrarian hedge fund trades. However, the drop in profitability may alternatively be due the inferences other investors make about that information from disclosure of the contrarian hedge fund advisers' 13Fs, which tend to be about 45 days after the end of the quarter.

Note that Table 8 Panel B shows no evidence or pattern of performance, seasonal or otherwise, for the momentum hedge fund advisers. While July exhibits a significantly positive buy-minus-sell alpha spread, this could be due to random chance, stemming from the multiple comparison. Likewise, analogous results for the other groups of funds show no evidence of performance. In addition, the quarter-end performance of hedge fund advisers is not consistent with patterns associated with leaning for the tape practices described by Carhart, Kaniel, Musto, and Reed (2002).²⁷ In sum, Table 8 is consistent with our paper's story of contrarian hedge fund advisers, but not the advisers of other groups of funds, possessing superior stock selection ability.

4.4. Fund Trades and Short Interest

Hedge fund managers frequently establish short positions in stocks, typically when they possess negative information about a company or the economic climate it operates in. The documented alphas of the long positions of hedge funds are unaffected by ignoring the short positions. Nev-

²⁶In unreported work using daily data, we find that the positive daily alphas of contrarian hedge funds are fairly evenly distributed throughout the first two months of the quarter.

²⁷Leaning for the tape involves trading on the last day of the quarter, which may allow fund managers to shift performance between periods.

ertheless, the overall portrait of hedge fund manager performance begs for some analysis of short positions. For example, Panel C of Table 1 indicates that the stocks held by hedge funds tend to have higher short interest ratios—the percentage of outstanding shares that are shorted. If the aggregate superior performance of hedge fund advisers from stock purchases comes at the expense of inferior performance by other hedge funds’ short sales in the same stock, focusing on long positions alone paints an incomplete picture of hedge fund managers’ skills.

While fund advisers are not required to disclose their short positions, we are able to analyze the monthly aggregate short interest for each stock with data provided by NYSE, AMEX, and NASDAQ.²⁸ The short-sale prohibitions and limitations imposed on most pension, endowment, and mutual fund advisers, as well as retail investors, greatly diminish the influence of non-hedge fund market participants on reported aggregate short sales.²⁹ Thus, it seems likely that these observed aggregate short positions are largely attributable to hedge funds and that hedge funds are relatively more pessimistic about the future returns of stocks with large aggregate short positions. In this vein, the extant literature on short sales and returns is consistent with our earlier conclusion that hedge fund managers are better performers. Several studies find significant negative relationships between a stock’s short-interest ratio (SIR) and its subsequent return. For example, [Boehmer, Huszar, and Jordan \(2010\)](#) find that stocks with low SIRs experience significant positive abnormal returns, while [Desai, Ramesh, Thiagarajan, and Balachandran \(2002\)](#) find that stocks with high SIRs experience significant negative abnormal returns.

Performance of short positions Our findings are consistent with the research mentioned above. We study short sales by first sorting stocks into two subsamples based on their end-of-prior-month SIR: below vs. above the month’s median SIR. Using the approach in Tables 5-7, we then classify each subsample’s stocks into equally weighted portfolios, based on whether the adviser type had purchased or sold the stock in the preceding reporting quarter. Table 9 Panel A reports that all of the buy and sell portfolios of below-median SIR stocks have CS alphas (controlling for size, BM, momentum, and 1-month past returns) that are 18-29 bpm larger than their above-median counterparts. This spread

²⁸For months prior to September 2007, there is monthly reporting of short interest as of the 15th of the month or the first trading day prior to the 15th. Since September 2007, short interest is reported bi-weekly. Our short interest time series uses the last day in the month with reported data.

²⁹For example, [Almazan, Brown, Carlson, and Chapman \(2004\)](#) find that about 70% of U.S. mutual funds are prohibited from short selling under their investment policy statements, and only 3% of mutual funds actually engage in short selling. Retail investors cannot hold short positions in qualified retirement accounts and face tax and carrying cost frictions that do not apply to their long positions.

is high enough to generate significantly positive alphas for low-SIR stocks that are either purchased or sold by momentum hedge fund advisers, despite there being no evidence that momentum hedge funds' trades add value on a relative basis. Thus, the opinion of short sellers about a stock is a factor affecting stock returns regardless of mutual and hedge fund trades.

While the SIR classification changes the alpha, it does not alter our conclusions about which groups enjoy good overall performance. For example, none of the buy-sell alpha spreads of mutual fund or of momentum hedge fund advisers in Panel A of Table 9 are significantly positive, irrespective of the SIR classification. The significantly positive buy-sell spreads of contrarian hedge funds with both SIR classifications reaffirm the superior performance of their fund managers. Indeed, the paper's finding regarding the skill of contrarian hedge fund advisers is only buttressed by the significantly negative alphas of the high-SIR stocks sold by the contrarian hedge fund advisers—stocks that short sellers are also pessimistic about.

Panel B of Table 9 examines whether the performance in Panel A differs based on concordance or disagreement between the groups of advisers. The diagonal entries of each panel's six 2x2 arrays represent stocks with concordance in the adviser pair's views; the off-diagonal entries represent stocks with disagreement. Sale and purchase concordance occur in each array's bottom right and top left cells, respectively. For each panel's three 2x2 arrays involving contrarian hedge funds (top third of the panels), the opposing adviser is worse off—whether buying or selling—when the trade of contrarian hedge funds in the stock are in the opposite direction.³⁰ Thus, Table 5-7's earlier finding of superior trades by contrarian hedge fund advisers is robust to whether the SIR is above or below the median. The costs of disagreeing with contrarian hedge fund advisers can be quite large. Selling low-SIR stocks that contrarian hedge funds buy reduces alpha by 23 to 58 bpm. For high-SIR stocks, positive alphas dominate the "sells" columns, except for the row where contrarian hedge funds also sell. This finding suggests that, even among high-SIR stocks, adviser types other than contrarian hedge fund advisers, and especially momentum mutual and hedge fund advisers, frequently sell future winners.

Thus, Table 9 enhances conclusions about the superior performance of contrarian hedge fund advisers. To the extent that short sales are representative of hedge fund positions, short sales improve the hedge fund manager performance inferred from studying their long positions.

³⁰The numbers reported in all three panels are similar to those that would be reported in DGTW-adjusted counterpart to the table, but omitted for brevity.

Style of short positions While 13F filings only report long equity positions, analysis of aggregate short interest reveals that hedge fund advisers' short positions also tend to be contrarian, reinforcing their style on the long side observed from 13F holdings. Specifically, Internet appendix Tables A10 and A11 show that past returns are positively correlated with changes in the aggregate short-interest ratio (largely driven by hedge funds). This covariance between aggregate stock-level short interest changes and stock returns is the short sales analog of the L0M style measure for the long side. Here, a positive number indicates a contrarian investment style because high past returns tend to be followed by an increase in short interest. The tables' results indicate that aggregate shorts tend to follow a contrarian strategy with respect to current- and prior-month returns. Similar (unreported) covariances between quarterly short interest changes (or SIR levels) and current- and prior-quarter returns also indicate contrarianism. Moreover, since stocks sold by contrarian hedge fund advisers and bought by momentum hedge fund advisers have greater increases in SIR than stocks bought by the former and sold by the latter group, the tables show that contrarian hedge fund short sales are more likely to be executed by contrarian than momentum hedge fund advisers.

4.5. Robustness Tests

Our results are robust to alternative weight calculations, alternative formation periods, alternative return adjustments, excluding index funds, and controlling for fund characteristics.

Alternative weights A stock's weight may change without a trade in the stock when an adviser's portfolio trades other securities or even when prices of other stocks in a portfolio change. To assess whether some of our results are driven by weight changes that are unrelated to discretionary buy and sell decisions in a stock, we also compute 'real' buy ('real' sell) L0M and F1M measures as partial sums only over the stocks for which there is an increase (decrease) in *both* portfolio weight and number of shares held. These 'real' style and performance measures (not reported) are virtually identical to the baseline case in Panel A of Tables 2 and 3.

Alternative formation period The baseline L0M measure is a cross-sectional covariance between stocks' quarterly returns and the difference in portfolio weights between an advisers' end-of- and

beginning-of-quarter positions.³¹ To assess whether more distant returns influence advisers' trades, we compute an L1M measure (Lag 1 Momentum), which relates current quarter weight changes to the *previous* quarter's returns.³² Internet appendix Table A12 Panel A shows that the L1M measures are consistent in sign but smaller in absolute value than the L0M measures, indicating that the styles of both mutual and hedge fund advisers are influenced more by recent returns. The L0M (L1M) measures are 0.71 (0.18) for mutual fund and -0.34 (-0.18) for hedge fund managers, all significant.

As a Table 4 robustness check, Internet appendix Table A6 examines performance results for advisers classified with respect to the style measures based on the L1M alternative formation period. Fund manager classifications as momentum or contrarian traders are generally consistent across the two measures for contrarian hedge fund and momentum mutual fund advisers, although the two measures frequently disagree on the classification of momentum hedge fund advisers. In cases where the measures disagree, the classification based on the L0M measure yields the greatest dispersion in performance between momentum and contrarian hedge fund advisers.

Alternative return adjustments Our main performance measures combine two risk-adjustment approaches: 1) *self-benchmarking*—using the returns the fund would earn if it did not trade over the quarter, and 2) *characteristics benchmarking*—benchmarking against the returns of stocks with similar characteristics by subtracting the returns of a matched characteristics portfolio from the returns of each stock an adviser trades. For robustness, Panel B of Internet appendix Table A12 presents momentum (L0M) and performance (F1M) measures based on raw and excess returns (calculated by subtracting from each stock's month- t return the month- t return of the CRSP value-weighted index). The alternative L0M and F1M measures produce similar results to the baseline case for both mutual and hedge fund advisers. For example, the raw, excess, SzBm-, and DGTW-adjusted L0M measures for hedge funds are -0.31 , -0.30 , -0.34 , and -0.34 , respectively, while the

³¹Since fund adviser holdings are observed at quarter-ends while the actual trading occurs throughout the quarter, some trades may take place before the returns in the L0M measure are observed. As GTW (1995, p. 1091) argue, this would not bias the style measure, except in the unlikely case when portfolio revisions occur predominantly at the beginning of the quarter and are informative about subsequent returns. In that case, the hedge fund L0M measure would be biased upwards—making detection of the contrarian hedge fund manager style we identify less likely and any finding of contrarian behavior conservative.

³²This measure is the same as GTW's L1M measure, except for an adjustment to the weights. The weights in L1M differ from the weights in L0M (see equation 2) in that they use the price at the end of quarter $q - 2$ instead of the price at the end of quarter $q - 1$, to ensure that weights are unaffected by subsequent returns.

corresponding F1M measures are 0.27, 0.25, 0.28, and 0.28.³³ The performance similarity across benchmarks indicates that the stocks bought by mutual and hedge fund managers have similar characteristics to the stocks the same adviser types sell.

Excluding passive managers We define as passive an adviser who rebalances less than 5% of portfolio holdings in any given quarter. Using this criterion, about 5% of the sample’s mutual fund and almost no hedge fund advisers are passive. A non-zero turnover threshold is required because some funds are closet indexers; other true index funds track indexes with strategies that do not hold the full index but mimic its returns almost perfectly; and even funds that hold the full set of stocks in a value-weighted index exhibit changes in portfolio holdings due to index additions, deletions, and dividends. Moreover, some advisers may be primarily indexers, but oversee a few actively-managed funds. Advisers meeting our passive criterion may dilute the style and performance of the adviser category they belong to. Excluding passive advisers leads to slightly higher LOM measures for mutual fund managers, indicating more momentum trading overall. However, the overall comparison of mutual and hedge fund advisers’ styles and performance is unaffected by excluding passive managers.

Different subperiods Internet appendix Table A13 compares adviser performance in the first and second half of the sample period. With one exception, the “All” column shows that for each adviser type, performance is similar and statistically indistinguishable across the two subperiods. Momentum hedge fund advisers did better in the first half of the sample period although, in both subperiods, their performance does not significantly differ from zero. When adjusting for the premium of the momentum strategy, only contrarian hedge fund advisers show significant alpha—in both subperiods!

Internet appendix Table A14 compares performance in two sets of months stratified by whether the momentum factor (and hence a momentum strategy) performs well or poorly. For contrarian hedge and mutual fund advisers, results are similar across good- and bad-momentum months with contrarian hedge fund advisers (as before, Table 5) exhibiting superior performance and contrarian mutual fund advisers exhibiting insignificant performance. Momentum hedge and mutual fund advisers, which have insignificant performance overall (Table 5), show significant positive performance in the good-momentum months and significant negative performance in the bad-momentum months.

³³In unreported results, we further regressed the time series of the raw and characteristic-adjusted F1M performance measures in Panel B on quarterly returns of the Fama and French (1993) and Carhart (1997) factors. The resulting time series alphas are of similar magnitude and significance as the ones reported in Panel B.

Disaggregation of mutual and hedge fund holdings Since 13F reports are filed at the fund adviser level, they combine mutual and hedge fund holdings of advisers to both types of funds. This aggregation raises the possibility that within-adviser netting and offsetting of positions may mask the true dominant style and performance of hedge fund vs. mutual fund advisers. We ensured that no advisers classified as hedge fund advisers manage mutual fund money. However, hedge fund capital represents a small fraction of assets for 20% of institutions classified as mutual fund advisers (based on the fact that more than 50% of their regulatory AUM belongs to mutual funds). To examine whether these hybrid fund managers differ in their performance and style, we further subdivide mutual fund advisers into 458 pure mutual fund and 131 hybrid managers. These tests show that hybrid fund advisers, who manage both mutual funds and a limited number of hedge funds, behave like pure mutual fund advisers in their style and performance. Since the holdings reported by hybrid fund managers predominantly belong to mutual funds, pooling these managers with mutual fund advisers does not significantly affect the results reported in the paper. The style and performance results of these tests are reported in Internet appendix Tables A1 and A2, respectively.

To further address the aggregation issue, we also collected fund-level data on mutual fund holdings from the Thomson Reuters database (S12). This data captures the holdings of 1,635 mutual funds managed by 458 mutual fund advisers and 979 mutual funds managed alongside hedge funds. The fund-level measures based on this data reveal no significant performance and style difference between mutual funds managed separately and those managed alongside hedge funds. The results of these tests are reported in Internet appendix Table A3.

A key advantage of 13F data over S12 data is that it is immune to cross-subsidization.³⁴ If hybrid advisers subsidize their hedge funds at the expense of their mutual funds, the aggregate 13F portfolio shows no gain or loss. However, the similarity of the S12 performance results on mutual funds to those using the 13Fs suggests that even if we were able to utilize fund-level data, the amount of cross-subsidization between a hybrid adviser's hedge and mutual funds is unlikely to be detectable.

³⁴See Chaudhuri, Ivković, and Trzcinka (2018) for an excellent discussion of cross-subsidization among U.S. equity products managed by institutional asset management firms.

5. Conclusion

Prior literature documents that mutual fund managers tend to follow momentum strategies, and their style accounts for any abnormal returns they generate. Using a database constructed from mandatory reporting of advisers' holdings of stocks, we find that the style and performance of hedge fund managers are remarkably different from those of mutual funds. Most hedge funds are contrarian, largely due to a strong tendency to buy recent losers. The losing stocks acquired by hedge funds do well over the subsequent quarter—earning significant alphas unaffected by whether alpha controls for the return-depressing effects of being a losing stock. It is only contrarian hedge fund managers who exhibit superior performance and performance persistence, outperforming both momentum hedge funds and risk-matched benchmarks. The superior performance of contrarian hedge funds is due to their managers' ability to identify underpriced stocks among losers that other investors avoid, rather than following a naïve rule-based strategy. In addition to adviser-level analysis, our findings are confirmed by a stock-level analysis. The highest alpha stocks are sold to contrarian hedge fund managers by momentum mutual fund managers.

Why do mutual fund managers perform poorly selling stocks that contrarian hedge fund managers buy (and vice versa)? One hypothesis is that mutual funds are subject to a number of regulations that constrain their trading behavior, offering profitable trading opportunities for their less constrained counterparts. For example, U.S. mutual funds are required to provide daily liquidity to their shareholders, which may lead to “fire sales” driven by large investor redemptions. [Chen, Hanson, Hong, and Stein \(2008\)](#) show that hedge fund returns are higher in months when the mutual fund sector is in distress and conclude that hedge funds engage in front-running to exploit the predictable trades of mutual funds. We also find that hedge fund managers' contrarian trading (buying of losers) increased during 2008-2009, while mutual fund advisers' selling of losers intensified. Moreover, mutual fund managers cater to retail investors' demand for ‘fashionable’ stocks, which may provide trading opportunities for contrarian hedge fund advisers.³⁵ In contrast, hedge funds are relatively less regulated, have more control over fund flows (e.g. through lock-up periods and limited windows for redemption), do not guarantee daily liquidity, and their investor base consists of institutional

³⁵[Massa, Simonov, and Yan \(2012\)](#) find international evidence that hedge funds perform better where mutual funds ownership is higher and argue that hedge funds take advantage of mutual funds' constraints. Moreover, studying aggregate mutual and hedge fund flows and aggregate cross-sectional mispricing, [Akbas, Armstrong, Sorescu, and Subrahmanyam \(2015\)](#) find that dumb money (retail investors/mutual funds) tends to exacerbate mispricing while smart money (hedge funds) attenuates it.

investors and high net-worth individuals with possibly longer-term investment horizons. Which of these hedge fund characteristics plays the most important role in explaining the superior performance of contrarian hedge fund advisers remains a question for future research.

The strong level of talent observed among contrarian hedge fund advisers is a bit surprising. Berk and Green's (2004)'s model of competition among advisers with scarce talent is characterized by decreasing returns per dollar invested. Such decreasing returns to scale should drive managerial rents per dollar invested to zero, as the most talented managers take on larger funds in equilibrium. Since we look at pre-expense alphas, not net returns that investors can achieve, our observation could still be consistent with Berk and Green (2004). Moreover, some models predict that positive pre-expense alphas are more likely with a contrarian investment philosophy. For example, in the rational expectations model of Watanabe (2008), investors with more precise private signals are contrarians, relying more on their private signals to detect when stock prices overshoot fair value, while less informed agents are trend followers, relying more on the public signal of price movements to form trades. In Makarov and Plantin (2015), unskilled managers use strategies (like momentum) that generate 'fake' alpha to temporarily manipulate investors' perception of their skills. Thus, while our findings are new, there is an underlying theoretical literature that it is consistent with and that ultimately, might be developed further with more careful study of our results.

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Table 1. Descriptive statistics

Panels A and B report the number of observations (unique fund advisers and adviser-quarter observations) and the average fund characteristics for mutual fund advisers, hedge fund advisers, and other institutional investors. Turnover is measured as the average of the dollar values of buys and sells over a quarter compared to the initial portfolio value: $\text{turnover} = (\$ \text{buys}_q + \$ \text{sells}_q) / (2 \times \$ \text{holdings}_{q-1})$. Panel C reports the time-series means of cross-sectional average characteristics of stocks held by the different types of investment adviser, weighted by the fraction of shares that the adviser holds. Bid-ask spread is the difference between the ask and bid prices from the monthly CRSP files as a percentage of the closing mid-price; analysts' forecast dispersion is the standard deviation in analysts' EPS forecasts standardized by the absolute value of the consensus forecast; SUE (standardized unexpected earnings) is the current quarter reported EPS minus the EPS four quarters prior, divided by the standard deviation of EPS changes over the last eight quarters. Panel D displays the percentage of stocks held by each type of adviser that are also held by other advisers. The sample period is 60 quarters long—from January 1998 to December 2012.

Characteristic	Adviser type		
	Mutual funds	Hedge funds	Other
<i>Panel A. Observations</i>			
Total unique fund advisers	589	1,342	2,894
Average fund adviser observations per quarter	354	484	1,230
Total adviser-quarter observations	21,219	29,050	73,821
Average life of adviser in sample (in quarters)	36.0	21.6	25.5
<i>Panel B. Average characteristics by fund adviser</i>			
Number of stocks per fund adviser	374	128	233
Stock holdings (\$millions)	11,288	864	3,103
Dollar holdings per stock (\$millions)	30.42	6.80	13.23
Turnover (%/quarter)	19.7	39.2	15.7
<i>Panel C. Share holdings-weighted average stock characteristics</i>			
Shares held/shares outstanding (%)	23.12	4.85	17.23
Price (\$)	33.84	26.95	35.61
Return (% per month)	0.65	0.90	0.60
Market capitalization (\$billions)	41.02	22.15	51.93
Book-to-market ratio	0.57	0.68	0.54
Trading volume (\$millions)	2.03	1.59	2.28
Bid-Ask Spread (% of price)	0.36	0.38	0.34
S&P long-term issuer credit rating	A–	BBB	A–
Volatility (% per month, $\sqrt{\sum r_{daily}^2} \times 100$)	11.01	12.29	10.51
Analyst coverage (# of analysts)	17.53	14.18	18.22
Analyst dispersion	0.14	0.21	0.12
SUE	0.85	0.61	0.92
Common dividends (\$/share)	0.62	0.38	0.73
Short interest (% of shares outstanding)	3.24	4.58	2.86
Institutional ownership (% of shares outstanding)	69.96	72.18	67.43
<i>Panel D. Stock holdings overlap across fund advisers</i>			
Mutual fund advisers	100.00	76.32	85.31
Hedge fund advisers	99.23	100.00	92.24
Other advisers	99.09	82.40	100.00

Table 2. Style measures

We compute the overall, buy, and sell $L0M_{iq}$ style measures for each fund adviser in each quarter as in equations (1), (8), and (9), respectively. Panel A presents, for each adviser type, the time-series mean of the quarterly average style measures, calculated as in equation (4), over the January 1998 to December 2012 period. Time-series t -statistics, calculated as in equation (5), are presented in parentheses (bold if significant at the 5% level). The percentage of positive quarterly measures is provided in brackets. Panel A also presents the cross-sectional mean of the adviser-level average style measures, as in equation (7), as well as the number and the proportion of advisers that are momentum ($L0M > 0$) and contrarian ($L0M < 0$) traders. For Panel B, we sort advisers into momentum and contrarian groups based on the sign of their $L0M_i$ style measure during the first half of the sample period (January 1998 to June 2005), and report for each group the average style measure $L0M_i$ and the proportion of advisers that are contrarian based on $L0M_i$ during the second half of the sample period (July 2005 to December 2012).

Panel A: Quarterly style measures

Measure	Adviser type					
	Mutual fund			Hedge fund		
	All	Buys	Sells	All	Buys	Sells
<i>Time-series average of adviser-level style measures</i>						
L0M	0.71 (14.88) [1.00]	0.39 (5.94) [0.87]	0.32 (5.66) [0.82]	-0.34 (- 3.62) [0.30]	-0.37 (- 2.43) [0.45]	0.03 (0.29) [0.48]
<i>Cross-sectional average of adviser-level style measures</i>						
L0M	1.07	0.47	0.59	-0.70	-0.66	-0.04
<i>Distribution of momentum ($L0M > 0$) and contrarian ($L0M < 0$) traders</i>						
Momentum traders (#)	396	352	406	475	444	644
Contrarians (#)	193	237	183	867	898	698
Proportion of contrarians	0.33	0.40	0.31	0.65	0.67	0.52

Panel B: Persistence of investment style

Style during first subperiod	Style during second subperiod					
	Average L0M			Proportion of contrarians ($L0M < 0$)		
	MF	HF	Other	MF	HF	Other
Contrarians ($L0M < 0$)	-0.33	-1.67	-0.37	0.71	0.79	0.63
Momentum traders ($L0M > 0$)	1.04	1.06	0.61	0.22	0.36	0.22

Table 3. Performance measures

We compute $F1M_{iq}^{SzBm}$ and $F1M_{iq}^{DGTW}$ performance measures for each fund adviser and each quarter as in equation (11) using size-, book-to-market- (and momentum-) adjusted returns. Panel A presents the time-series mean of the quarterly average performance measures for each adviser type over the January 1998 to December 2012 period. Time-series t -statistics are in parentheses (bold if significant at the 5% level). The percentage of positive quarterly measures is provided in brackets. Panel A also presents the cross-sectional mean of the fund-level average performance measures. For Panel B, we sort fund advisers into underperformers/outperformers based on their average $F1M^{SzBm}$ ($F1M^{DGTW}$) during the first half of the sample period (January 1998 to June 2005), and report their average $F1M^{SzBm}$ ($F1M^{DGTW}$) performance during the second half of the sample period (July 2005 to December 2012).

Panel A: Quarterly performance measures

Measure	Adviser type					
	Mutual fund			Hedge fund		
	All	Buys	Sells	All	Buys	Sells
<i>Time-series average of adviser-level performance measures</i>						
$F1M^{SzBm}$	0.13 (2.67) [0.65]	0.10 (1.71) [0.62]	0.03 (0.82) [0.52]	0.27 (3.61) [0.65]	0.28 (2.57) [0.62]	-0.01 (-0.12) [0.45]
$F1M^{DGTW}$	0.02 (0.51) [0.58]	0.04 (0.99) [0.58]	-0.02 (-0.75) [0.47]	0.25 (3.72) [0.68]	0.26 (3.17) [0.65]	-0.01 (-0.15) [0.47]
<i>Cross-sectional average of adviser-level performance measures</i>						
$F1M^{SzBm}$	0.12	-0.00	0.12	0.46	0.27	0.19
$F1M^{DGTW}$	-0.08	-0.06	-0.02	0.40	0.26	0.15

Panel B: Persistence of performance

Performance during first subperiod	Performance during second subperiod					
	$F1M^{SzBm}$			$F1M^{DGTW}$		
	MF	HF	Other	MF	HF	Other
Underperformers ($F1M < 0$)	-0.018 (-0.22)	0.048 (0.42)	0.024 (0.93)	-0.005 (-0.08)	0.176 (1.78)	-0.002 (-0.10)
Number of advisers	137	217	701	150	234	740
Average life (qtrs) 1 st half	28	16	20	27	16	20
Average life (qtrs) 2 nd half	24	21	22	24	21	22
% of funds still in 2 nd half	71%	76%	72%	68%	78%	72%
Outperformers ($F1M > 0$)	0.055 (1.54)	0.428 (3.65)	-0.025 (-0.54)	-0.021 (-0.70)	0.327 (2.98)	-0.036 (-0.78)
Number of advisers	223	322	639	210	305	600
Average life (qtrs) 1 st half	27	16	19	27	16	20
Average life (qtrs) 2 nd half	24	22	22	24	22	23
% of funds still in 2 nd half	71%	79%	70%	72%	78%	70%

Table 4. Performance of momentum and contrarian fund advisers

The table presents the time-series (Panel A) and cross-sectional (Panel B) average performance measures $F1M^{SzBm}$ and $F1M^{DGTW}$ for contrarian and momentum traders for both mutual and hedge fund advisers. The SzBm (DGTW) label indicates that the measure uses returns adjusted for size and book-to-market (and momentum). Momentum (contrarian) advisers are those with a positive (negative) average performance measure $L0M_i$ over the life of the adviser. There are 396 (193) momentum (contrarian) mutual fund advisers and 475 (867) momentum (contrarian) hedge fund advisers. Time-series t -statistics are presented in parentheses (bold if significant at the 5% level).

Measure	Adviser type					
	Mutual fund			Hedge fund		
	All	Buys	Sells	All	Buys	Sells
<i>Panel A: Time-series average of adviser-level quarterly performance measures</i>						
Contrarian fund advisers ($L0M < 0$)						
$F1M^{SzBm}$	-0.02 (-0.55)	0.03 (0.83)	-0.04 (-1.34)	0.35 (3.99)	0.34 (2.85)	0.01 (0.12)
$F1M^{DGTW}$	0.01 (0.31)	0.04 (1.42)	-0.03 (-1.42)	0.34 (4.09)	0.33 (3.52)	0.01 (0.21)
Momentum fund advisers ($L0M > 0$)						
$F1M^{SzBm}$	0.22 (2.91)	0.14 (1.58)	0.08 (1.26)	0.10 (0.95)	0.14 (0.89)	-0.04 (-0.33)
$F1M^{DGTW}$	0.02 (0.42)	0.04 (0.68)	-0.02 (-0.41)	0.08 (0.86)	0.13 (1.05)	-0.05 (-0.53)
<i>Panel B: Cross-sectional average of adviser-level performance measures</i>						
Contrarian funds ($L0M < 0$)						
$F1M^{SzBm}$	-0.11	-0.07	-0.04	0.65	0.44	0.21
$F1M^{DGTW}$	-0.04	-0.03	-0.01	0.60	0.41	0.19
Momentum funds ($L0M > 0$)						
$F1M^{SzBm}$	0.23	0.03	0.20	0.12	-0.03	0.15
$F1M^{DGTW}$	-0.10	-0.08	-0.03	0.04	-0.03	0.07

Table 5. DGTW-adjusted returns sorted on aggregate trades

Each month, stocks are included in equally weighted monthly buy (sell) portfolios if their aggregate holdings by a particular adviser type increased (decreased) over the previous quarter. The table reports time-series averages and sample *t*-statistics of DGTW-adjusted monthly portfolio returns. Panel A presents buy, sell, and buy-minus-sell portfolio returns by fund adviser category and by style. Panel B summarizes portfolio returns based on the interaction between the trades of mutual and hedge fund advisers. Stocks that are bought by contrarian hedge fund advisers and sold by momentum mutual fund advisers (1st row and 2nd column) are stocks for which the former adviser category's aggregate portfolio weights increased and the latter category's weights decrease (after adjustment for the passive portfolio weight change due to stock price changes). The equal weighted portfolio return in the first row and second column of Panel B thus represents the average return of stocks that effectively are sold by momentum mutual fund advisers to contrarian hedge fund advisers. The sample period is from January 1998 to December 2012.

Panel A. Buys and sells by fund adviser type

Adviser type	Buys	Sells	Buys–Sells
Mutual fund	0.004 (0.06)	0.207 (3.32)	−0.203 (− 2.41)
Hedge fund	0.187 (3.26)	0.004 (0.10)	0.183 (3.07)
Contrarian hedge fund	0.224 (4.70)	−0.063 (−1.41)	0.287 (5.84)
Momentum hedge fund	0.145 (1.93)	0.146 (3.25)	−0.001 (−0.02)
Contrarian mutual fund	0.081 (1.95)	0.135 (2.61)	−0.054 (−1.03)
Momentum mutual fund	0.029 (0.41)	0.196 (3.17)	−0.166 (−1.63)

Panel B. Buy-Sell interactions by mutual fund-hedge fund adviser style

	Momentum MF		Contrarian MF		Momentum HF	
	Buys	Sells	Buys	Sells	Buys	Sells
Contrarian hedge fund buys	0.122 (1.52)	0.353 (4.57)	0.222 (4.41)	0.229 (3.30)	0.242 (3.43)	0.311 (5.34)
Contrarian hedge fund sells	−0.048 (−0.62)	−0.054 (−0.71)	−0.064 (−1.26)	−0.015 (−0.24)	−0.002 (−0.03)	−0.056 (−0.91)
Momentum hedge fund buys	0.139 (1.19)	0.154 (2.56)	0.143 (1.95)	0.177 (1.93)		
Momentum hedge fund sells	0.053 (0.83)	0.281 (3.03)	0.166 (2.88)	0.142 (2.45)		
Contrarian mutual fund buys	0.025 (0.36)	0.161 (2.86)				
Contrarian mutual fund sells	0.082 (0.98)	0.245 (2.56)				

Table 6. CS-adjusted returns sorted on aggregate fund trades

Each month, stocks are included in equally weighted monthly buy (sell) portfolios if their aggregate holdings by a particular adviser type increased (decreased) over the previous quarter. The table reports time-series averages and sample *t*-statistics of monthly portfolio returns which were adjusted in a cross-sectional regression with log(size), log(BM), cumulative past-6-month return, and past-month return (as a proxy for liquidity provision). Panel A presents buy, sell, and buy-minus-sell portfolio returns by adviser category and by style. Panel B summarizes portfolio returns based on the interaction between the trades of mutual and hedge fund advisers. The sample period is from January 1998 to December 2012.

Panel A. Buys and sells by fund adviser

Adviser type	Buys	Sells	Buys–Sells
Mutual fund	0.007 (0.26)	0.167 (5.24)	−0.160 (−3.22)
Hedge fund	0.182 (5.47)	−0.015 (−0.37)	0.196 (4.12)
Contrarian hedge fund	0.236 (6.88)	−0.116 (−3.15)	0.352 (7.73)
Momentum hedge fund	0.112 (2.59)	0.131 (2.66)	−0.018 (−0.34)
Contrarian mutual fund	0.105 (3.19)	0.079 (2.42)	0.027 (0.54)
Momentum mutual fund	0.019 (0.60)	0.176 (4.61)	−0.158 (−2.85)

Panel B. Buy-Sell interactions by mutual fund-hedge fund adviser style

	Momentum MF		Contrarian MF		Momentum HF	
	Buys	Sells	Buys	Sells	Buys	Sells
Contrarian hedge fund buys	0.129 (2.79)	0.356 (6.02)	0.247 (5.03)	0.236 (5.19)	0.210 (4.21)	0.323 (5.63)
Contrarian hedge fund sells	−0.112 (−2.19)	−0.072 (−1.11)	−0.065 (−1.34)	−0.137 (−2.59)	−0.060 (−1.01)	−0.102 (−1.64)
Momentum hedge fund buys	0.091 (1.54)	0.164 (2.69)	0.131 (2.48)	0.111 (1.94)		
Momentum hedge fund sells	0.052 (0.87)	0.244 (3.25)	0.180 (3.05)	0.093 (1.53)		
Contrarian mutual fund buys	0.044 (0.93)	0.189 (3.77)				
Contrarian mutual fund sells	0.034 (0.73)	0.205 (2.98)				

Table 7. Three-attribute CS-adjusted returns sorted on aggregate fund trades

Each month, stocks are included in equally weighted monthly buy (sell) portfolios if their aggregate holdings by a particular adviser type increased (decreased) over the previous quarter. The table reports time-series averages and sample *t*-statistics of monthly portfolio returns which were adjusted in a cross-sectional regression with $\log(\text{size})$, $\log(\text{BM})$, and cumulative past 6 month return. Panel A presents buy, sell, and buy-minus-sell portfolio returns by adviser category and by style. Panel B summarizes portfolio returns based on the interaction between the trades of mutual and hedge fund advisers. The sample period is from January 1998 to December 2012.

Panel A. Buys and sells by fund adviser type

Adviser type	Buys	Sells	Buys–Sells
Mutual fund	0.002 (0.08)	0.188 (5.48)	−0.186 (−3.50)
Hedge fund	0.169 (5.13)	−0.005 (−0.11)	0.174 (3.67)
Contrarian hedge fund	0.242 (6.78)	−0.127 (−3.39)	0.369 (7.74)
Momentum hedge fund	0.080 (1.82)	0.149 (2.92)	−0.069 (−1.22)
Contrarian mutual fund	0.110 (3.24)	0.083 (2.49)	0.027 (0.54)
Momentum mutual fund	0.010 (0.29)	0.200 (4.97)	−0.191 (−3.27)

Panel B. Buy-Sell interactions by mutual fund-hedge fund style

	Momentum MF		Contrarian MF		Momentum HF	
	Buys	Sells	Buys	Sells	Buys	Sells
Contrarian hedge fund buys	0.116 (2.40)	0.377 (6.01)	0.255 (4.93)	0.238 (5.09)	0.189 (3.78)	0.354 (5.85)
Contrarian hedge fund sells	−0.138 (−2.61)	−0.061 (−0.92)	−0.080 (−1.62)	−0.155 (−2.85)	−0.109 (−1.78)	−0.105 (−1.67)
Momentum hedge fund buys	0.044 (0.72)	0.160 (2.60)	0.102 (1.93)	0.077 (1.29)		
Momentum hedge fund sells	0.051 (0.83)	0.279 (3.56)	0.197 (3.20)	0.112 (1.83)		
Contrarian mutual fund buys	0.030 (0.62)	0.213 (4.08)				
Contrarian mutual fund sells	0.025 (0.52)	0.220 (3.12)				

Table 8. Seasonality in hedge fund returns

Each month, stocks are included in equally weighted monthly buy (sell) portfolios if their aggregate holdings by a particular adviser type increased (decreased) over the previous quarter. The table reports time-series averages and sample t -statistics of monthly portfolio returns for each calendar month. The returns were adjusted each month in cross-sectional regressions with $\log(\text{size})$, $\log(\text{BM})$, cumulative past-6-month return, and past-month return. The table reports buy, sell, and buy-minus sell returns for contrarian hedge fund advisers. The sample period is from January 1998 to December 2012.

Portfolio	Buys	Sells	Buys–Sells
<i>Panel A. Contrarian hedge fund advisers</i>			
January	0.245 (2.11)	−0.238 (−2.05)	0.483 (6.31)
February	0.090 (0.95)	−0.082 (−0.67)	0.172 (1.13)
March	0.124 (1.23)	0.090 (0.77)	0.034 (0.20)
April	0.330 (1.90)	−0.078 (−0.60)	0.408 (2.30)
May	0.232 (1.86)	−0.256 (−3.00)	0.488 (2.74)
June	0.223 (2.53)	0.049 (0.44)	0.174 (1.43)
July	0.196 (1.47)	−0.343 (−1.88)	0.540 (2.97)
August	0.274 (1.80)	−0.219 (−1.82)	0.493 (4.65)
September	0.168 (2.40)	−0.090 (−0.58)	0.258 (1.53)
October	0.312 (2.65)	−0.266 (−1.95)	0.578 (2.94)
November	0.452 (4.46)	−0.123 (−1.05)	0.575 (4.19)
December	0.196 (1.42)	0.170 (2.44)	0.026 (0.18)
<i>Panel B. Momentum hedge fund advisers</i>			
January	−0.105 (−0.90)	−0.155 (−0.97)	0.050 (0.24)
February	−0.147 (−0.85)	0.213 (1.36)	−0.361 (−1.63)
March	0.083 (0.91)	0.436 (3.55)	−0.353 (−2.36)
April	0.373 (2.03)	0.147 (0.77)	0.226 (1.10)
May	−0.181 (−1.95)	−0.017 (−0.11)	−0.164 (−0.93)
June	0.243 (1.71)	0.292 (2.20)	−0.049 (−0.30)
July	0.236 (1.23)	−0.360 (−1.98)	0.595 (4.01)
August	−0.072 (−0.48)	0.107 (0.55)	−0.179 (−1.15)
September	0.128 (1.09)	0.151 (1.46)	−0.023 (−0.28)
October	0.174 (1.12)	0.103 (0.44)	0.070 (0.28)
November	0.220 (1.99)	0.163 (1.39)	0.058 (0.33)
December	0.414 (2.23)	0.488 (2.79)	−0.074 (−0.42)

Table 9. CS-adjusted returns by aggregate fund trades and short interest

Each month, stocks are included in equally weighted monthly buy (sell) portfolios if their aggregate holdings by a particular adviser type increased (decreased) over the previous quarter. In addition, stock are sorted in two groups (below- and above-median) according to their month-end short interest ratio (SIR). The table reports time-series averages and sample *t*-statistics for monthly portfolio returns (in percent) across fund advisers, adviser styles, and short interest, adjusted using cross-sectional regressions on size, BM, momentum, and past-month return. Panel A summarizes buy, sell, and buy-minus-sell portfolio returns (in percent) by adviser category within stocks with below- (above-) median SIR, respectively. Panel B summarizes portfolio returns based on the interaction between the trades of mutual and hedge fund advisers for stocks with low (high) SIR, respectively. The sample period is from January 1998 to December 2012.

Panel A. Buys and sells by fund adviser type and SIR

Adviser type	Below-median SIR			Above-median SIR		
	Buys	Sells	Buys–Sells	Buys	Sells	Buys–Sells
Mutual fund	0.239 (3.05)	0.349 (4.73)	−0.110 (−1.96)	−0.049 (−0.69)	0.172 (2.30)	−0.221 (− 3.26)
Hedge fund	0.398 (5.21)	0.221 (2.83)	0.177 (2.83)	0.133 (2.08)	−0.017 (−0.22)	0.150 (2.49)
Contrarian hedge fund	0.453 (5.55)	0.152 (1.97)	0.301 (4.07)	0.212 (3.17)	−0.136 (−1.90)	0.347 (6.11)
Momentum hedge fund	0.350 (4.16)	0.381 (4.24)	−0.031 (−0.44)	0.064 (0.99)	0.126 (1.65)	−0.062 (−0.97)
Contrarian mutual fund	0.321 (4.35)	0.286 (3.82)	0.034 (0.52)	0.063 (0.89)	0.061 (0.89)	0.002 (0.04)
Momentum mutual fund	0.246 (3.23)	0.367 (4.94)	−0.121 (−1.93)	−0.032 (−0.42)	0.165 (2.12)	−0.197 (− 2.53)

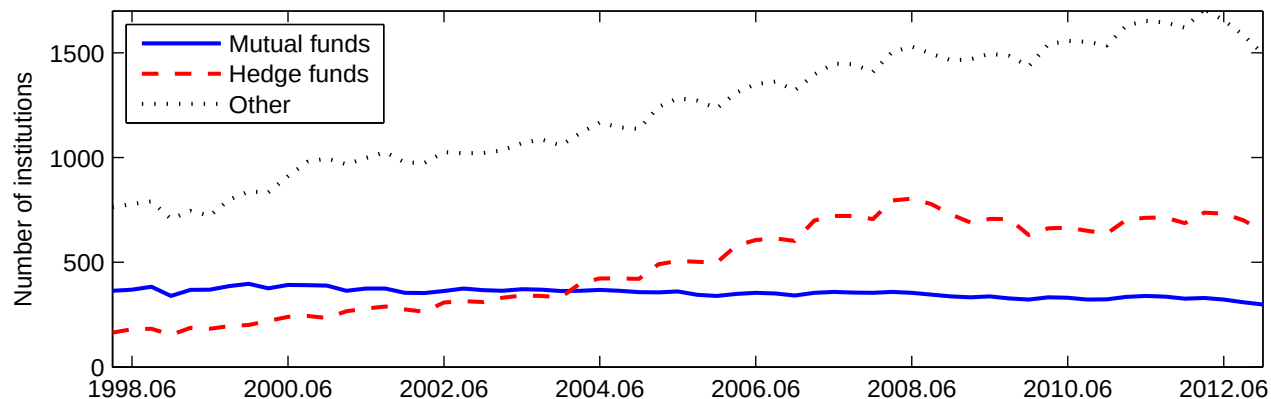
Panel B. Buy-Sell interactions by fund adviser style and SIR

	Momentum MF		Contrarian MF		Momentum HF	
	Buys	Sells	Buys	Sells	Buys	Sells
<i>Below-median short interest</i>						
Contrarian hedge fund buys	0.475 (5.29)	0.448 (4.83)	0.507 (5.38)	0.381 (3.91)	0.515 (4.96)	0.577 (5.43)
Contrarian hedge fund sells	0.106 (1.17)	0.296 (3.02)	0.247 (2.75)	0.109 (1.14)	0.187 (1.88)	0.186 (1.62)
Momentum hedge fund buys	0.357 (3.83)	0.435 (3.76)	0.418 (4.45)	0.348 (3.23)		
Momentum hedge fund sells	0.397 (3.63)	0.429 (4.15)	0.432 (3.100)	0.289 (2.67)		
Contrarian mutual fund buys	0.327 (4.14)	0.361 (4.22)				
Contrarian mutual fund sells	0.241 (2.77)	0.428 (4.20)				
<i>Above-median short interest</i>						
Contrarian hedge fund buys	0.035 (0.42)	0.402 (4.70)	0.212 (2.73)	0.233 (3.13)	0.173 (2.52)	0.295 (3.48)
Contrarian hedge fund sells	−0.105 (−1.22)	−0.145 (−1.54)	−0.109 (−1.38)	−0.145 (−1.76)	−0.100 (−1.25)	−0.095 (−1.09)
Momentum hedge fund buys	0.040 (0.48)	0.104 (1.32)	0.083 (1.11)	0.055 (0.73)		
Momentum hedge fund sells	0.006 (0.07)	0.270 (2.59)	0.157 (1.89)	0.119 (1.34)		
Contrarian mutual fund buys	−0.045 (−0.52)	0.191 (2.31)				
Contrarian mutual fund sells	0.009 (0.11)	0.178 (1.76)				

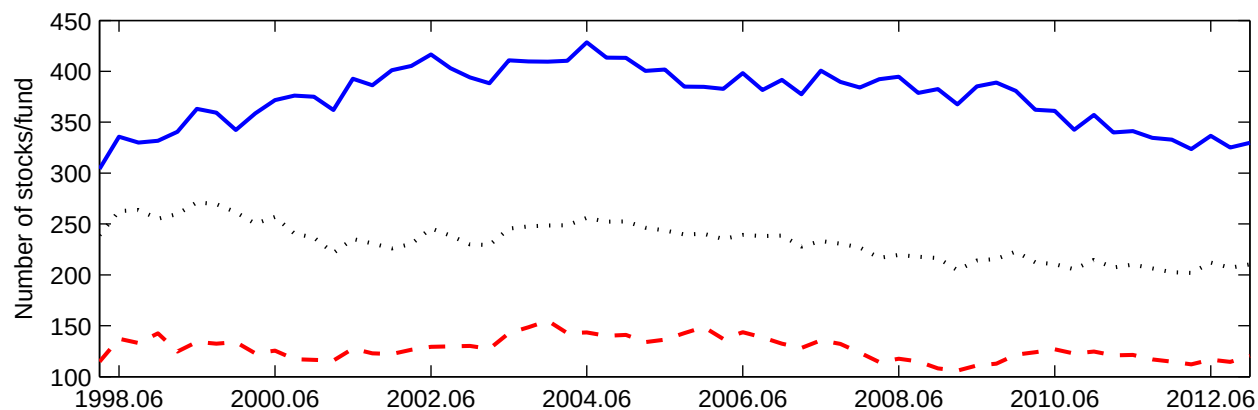
Figure 1. Time-series of institutional observations, stock holdings, and turnover

Plot A presents the number of mutual fund, hedge fund, and other institutional advisers in our sample over the 60-quarter period between January 1998 to December 2012. Plot B presents the number of stocks held by the average mutual and hedge fund adviser throughout the sample period. Plot C presents the turnover of the average mutual and hedge fund adviser, measured as $\text{turnover} = [(\$\text{buys}_q + \$\text{sell}_q) / (2 \times \$\text{holdings}_{q-1})]$.

Plot A: Number of fund advisers



Plot B: Number of stocks per fund adviser



Plot C: Turnover per quarter

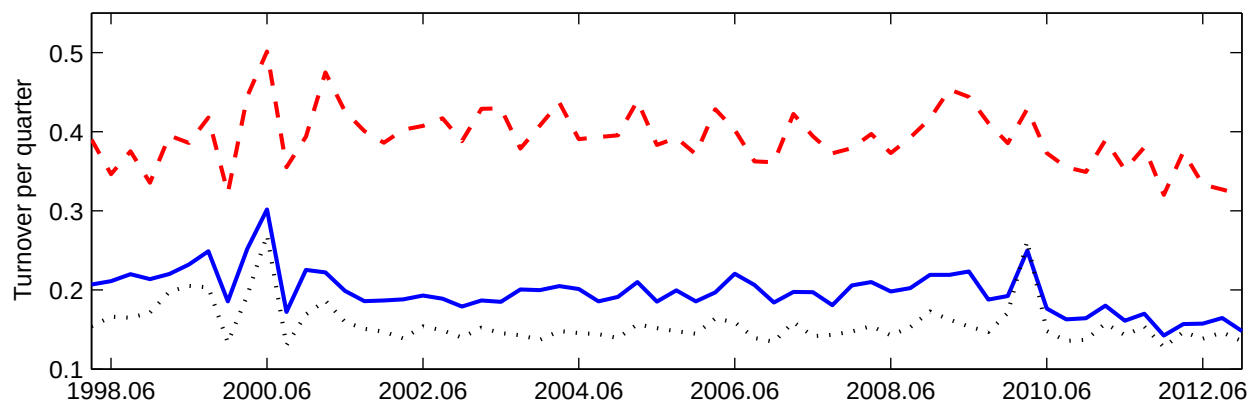
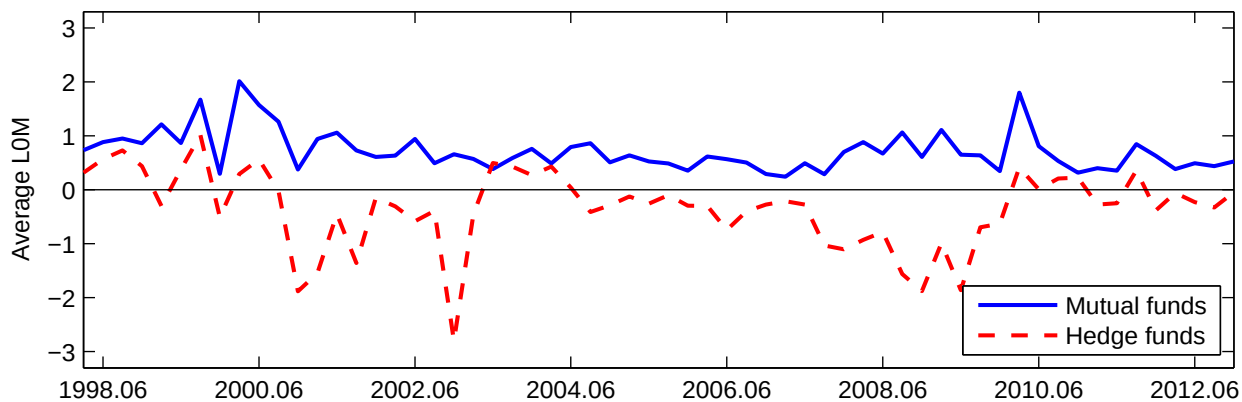


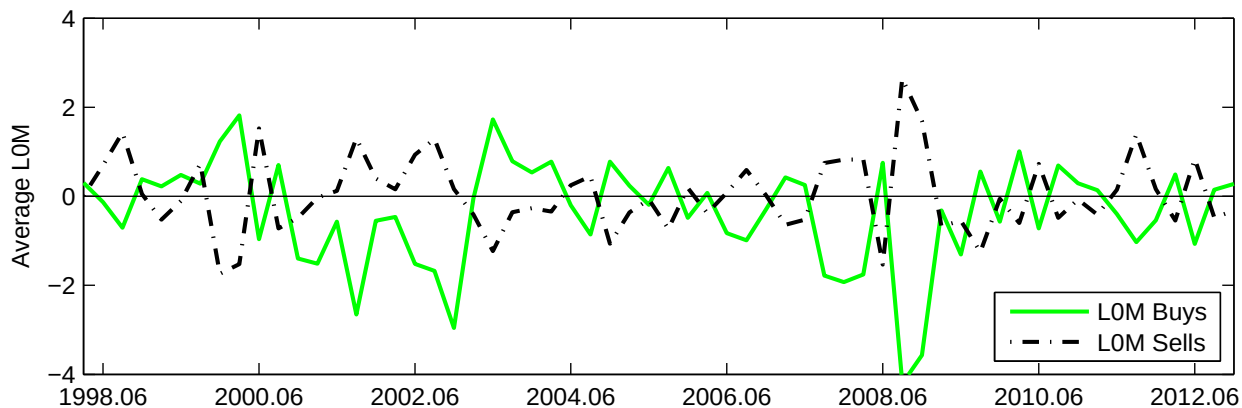
Figure 2. Time-series of style measures

Plot A presents the time-series of the average LOM style measure (equation 1) across mutual (hedge) fund advisers over the 60-quarter period between January 1998 to December 2012. Plot B (C) presents the contributions of buys and sells to the overall LOM style measure for hedge fund (mutual fund) advisers.

Plot A: Style measures (LOM) for mutual and hedge fund advisers



Plot B: Hedge fund advisers' buys and sells



Plot C: Mutual fund advisers' buys and sells

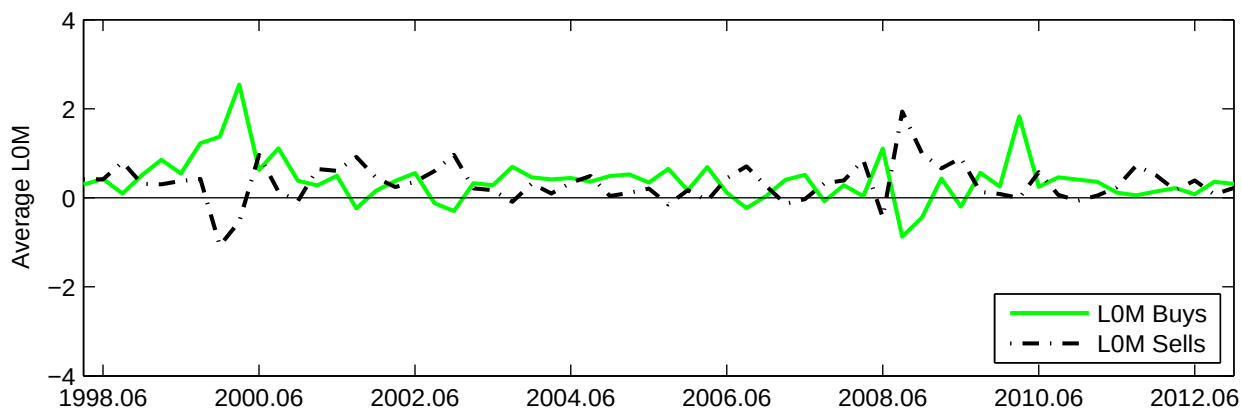
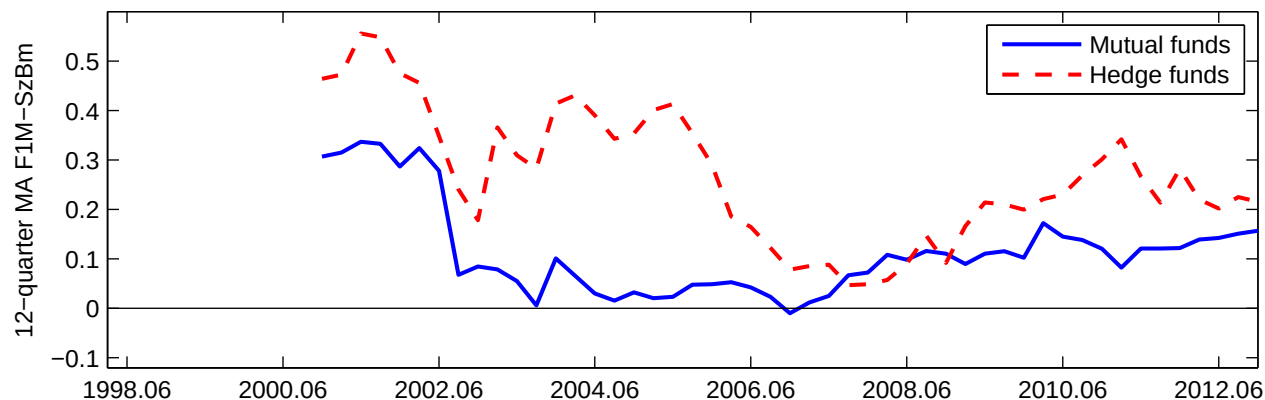


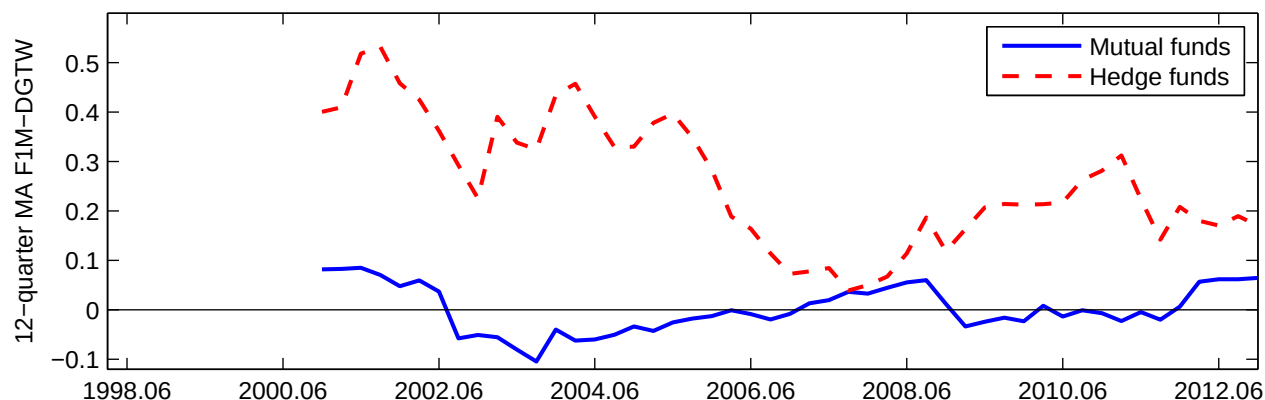
Figure 3. Time-series of performance measures

Plots A and B present the 12-quarter moving average of the performance measures (equation 11) for hedge fund and mutual fund advisers. The SzBm (DGTW) label indicates that the measure uses returns adjusted for size and book-to-market (Plot A) and momentum (Plot B). Plot C presents the contributions of buys and sells to the overall $F1M^{DGTW}$ performance measure for hedge fund advisers.

Plot A: Performance measure $F1M^{SzBm}$ for mutual and hedge fund advisers



Plot B: Performance measure $F1M^{DGTW}$ for mutual and hedge fund advisers



Plot C: The contribution of buys and sells to hedge fund performance based on the $F1M^{DGTW}$ measure

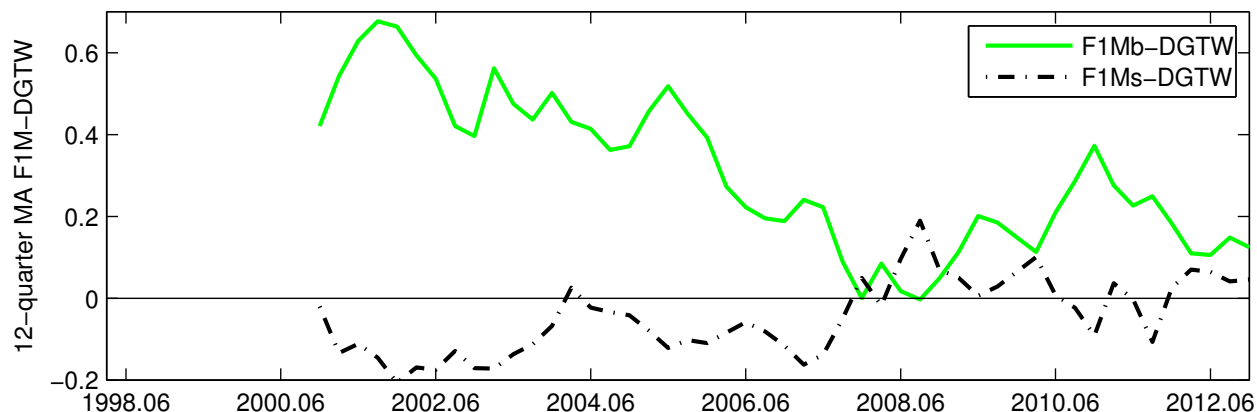
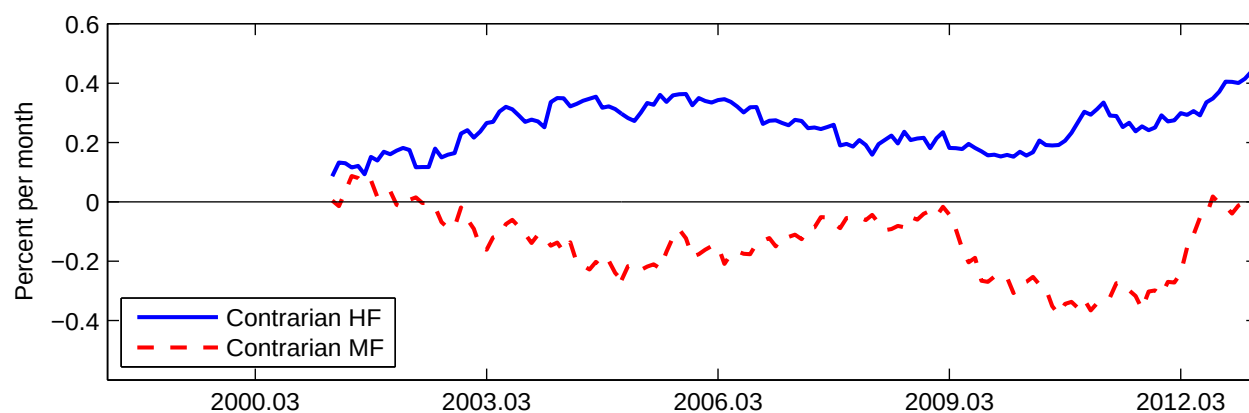


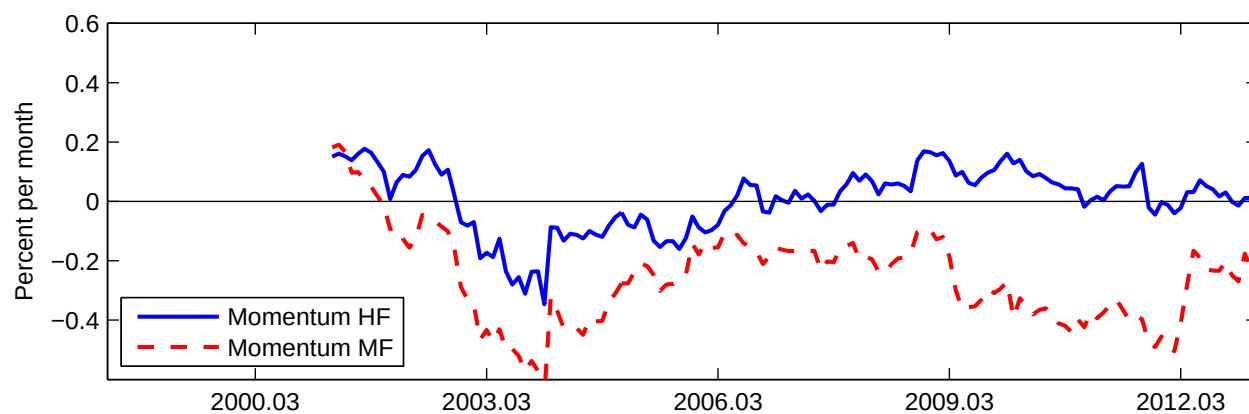
Figure 4. Time-series of DGTW-adjusted monthly portfolio returns

Each quarter, we sort stocks into buy and sell portfolios depending on whether a particular type of adviser increased or decreased their aggregate holdings of the stock over the previous quarter. For each type of fund, we take the difference between the equally weighted DGTW-adjusted monthly stock returns of their buys and sells. Plots A and B present the 36-month moving average of DGTW-adjusted buy–sell returns for portfolios based on the trades of contrarian and momentum fund advisers, respectively. Plot C compares contrarian and momentum hedge fund advisers.

Plot A: DGTW-adjusted portfolio returns for contrarian fund advisers



Plot B: DGTW-adjusted portfolio returns for momentum fund advisers



Plot C: DGTW-adjusted portfolio returns for contrarian and momentum hedge fund advisers

